

Lookout 2: Anomaly detection via leave-one-out kernel density estimation

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Abstract

We present an updated version of *lookout* – an algorithm for detecting anomalies using kernel density estimates with bandwidth based on Rips death diameters – with theoretical guarantees. The kernel density estimator for updated *lookout* is shown to be consistent, and the proposed multivariate scaling is robust and efficient. We show our updated algorithm performs better than the previous version on diverse examples.

Keywords: bandwidth selection, extreme value theory, generalized Pareto distribution minimum spanning tree, outlier detection, single-linkage clustering.

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1 Introduction

Anomaly detection is used in many application domains such as credit card fraud and scam detection (Kolupuri et al. 2025). Kandanaarachchi & Hyndman (2022) proposed the *lookout* algorithm for anomaly detection using a kernel density estimate with bandwidth based on persistence homology. Anomalies are identified by fitting a Generalized Pareto Distribution (GPD) to the negative logarithm of the smallest leave-one-out kernel density estimates, and declaring points with low GPD probabilities as anomalies.

In this paper, we explore some of the properties of the lookout algorithm, particularly the choice of bandwidth and what that choice implies for the kernel density estimates it produces. The bandwidth was originally expressed in terms of the death diameters of a Vietoris-Rips filtration, but those death diameters are exactly the edge lengths of the Euclidean minimum spanning tree of the data, and it is the minimum spanning tree that we work with in this paper. This allows us to analyse the asymptotic properties of the algorithm, and to propose a modified lookout algorithm that takes its bandwidth from a different feature of the same set of edge lengths. The modified algorithm improves the performance and robustness of the original algorithm, under a broader range of conditions.

First we introduce some background material, before describing both the original lookout algorithm, and our proposed modifications.

1.1 Anomalies and surprisals

There is no broadly accepted definition for an anomaly. Hawkins (1980) defines an anomaly or outlier as *an observation which deviates so much from other observations as to arouse suspicion it was generated by a different mechanism*. Barnett & Lewis (1978) define an anomaly or outlier as *an observation (or a subset of observations) which appears to be inconsistent with the remainder of that set of data*. Both these definitions agree that anomalies are rare and different from other observations. That can be applied by fitting some (conditional) probability distribution to the set of observations, and considering whether each observation is a likely random draw from the probability distribution. Hyndman & Frazier (2026) proposed a formalization of this concept, which we state in the following definition. Throughout, f is a probability density function on $\mathbb{R}^m$ with respect to Lebesgue measure. It is the density against which the observations are judged, and we do not assume that every observation is a draw from it: the anomalies

are precisely those that appear not to be.

Definition 1 (Anomalies). Given a set of observations $\mathbf{y}_i \in \mathbb{R}^m$ for $i \in \{1, \dots, n\}$, and a probability density function f , we define observation $\mathbf{y}_i$ as anomalous if

$$\mathbb{P}(f(\mathbf{Y}) \leq f(\mathbf{y}_i)) < \alpha, \quad (1)$$

where $\mathbf{Y}$ is a random vector with density f , and $\alpha > 0$ is a chosen threshold.

Definition 2 (Surprisals). Given a probability density function f , the *surprisal* of an observation $\mathbf{y}$ drawn from the density f is defined as $-\log f(\mathbf{y})$.

These are better known as “log scores” in statistics (Good 1952), but we prefer the term “surprisals” from information theory (Stone 2022), as it provides a measure of how surprising an observation is, given the probability density function f . Specifically, it takes large values for observations that are unlikely under f .

Let $S = -\log f(\mathbf{Y})$ be the random variable representing the surprisal of a random vector $\mathbf{Y}$ drawn from f , and let $G(s) = \mathbb{P}(S \leq s)$ be its distribution function. Then the surprisal probability of observation $\mathbf{y}_i$ is given by

$$p_i = 1 - G(s_i^-),$$

where $s_i = -\log f(\mathbf{y}_i)$, and $G(s_i^-) = \lim_{t \uparrow s_i} G(t)$ is the left limit of G at s_i . An observation is an anomaly if and only if $p_i < \alpha$.

1.2 Extreme value theory

The lookout algorithm determines which surprisals are unusually large by using a peaks-over-threshold approach (Pickands 1975, Coles 2001), and fitting a Generalized Pareto distribution (GPD) to the upper tail of the observed surprisals. That is,

$$\mathbb{P}(S \leq s \mid S > u) = \frac{G(s) - G(u)}{1 - G(u)} \approx 1 - (1 + \xi(s - u)/\sigma_u)^{-1/\xi}, \quad 1 + \xi(s - u)/\sigma_u > 0, \quad (2)$$

where $s > u$ and u is a threshold chosen to be large enough that the conditional distribution of the excess $S - u$ over the threshold is well approximated by a GPD, $\sigma_u > 0$ is a scale parameter depending

on the threshold, $\xi \in \mathbb{R}$ is a shape parameter that does not, and $\xi = 0$ is read as the limiting case $1 - \exp(-(s - u)/\sigma_u)$.

The distribution in (2) is unbounded above when $\xi \geq 0$, and has finite upper endpoint $u + \sigma_u/|\xi|$ when $\xi < 0$, so a quantity that is bounded above cannot have a positive shape parameter. We use this in Section 4 to constrain the fitted GPD.

1.3 Kernel density estimation

Given a set of observations $\mathbf{y}_i \in \mathbb{R}^m$ for $i \in \{1, \dots, n\}$, a kernel density estimate is defined as (Chacón & Duong 2018)

$$\hat{f}(\mathbf{y}) = \frac{1}{n} \sum_{i=1}^n |\mathbf{H}|^{-1/2} K(\mathbf{H}^{-1/2}(\mathbf{y} - \mathbf{y}_i)), \quad (3)$$

where K is a kernel function and $\mathbf{H}$ is a symmetric $m \times m$ positive-definite bandwidth matrix. When the observations are n independent draws from f , the accuracy of $\hat{f}$ is conventionally measured by the mean integrated squared error (MISE),

$$\text{MISE}(\hat{f}) = \mathbb{E} \left[\int (\hat{f}(\mathbf{u}) - f(\mathbf{u}))^2 d\mathbf{u} \right], \quad (4)$$

where the expectation is taken over that sample, and we say that $\hat{f}$ is a *consistent estimator* of f if $\text{MISE}(\hat{f}) \rightarrow 0$ as $n \rightarrow \infty$. Consistency holds under the following conditions on the density, the kernel and the bandwidth matrix (Chacón & Duong 2018, pp. 31–37).

A1 The density function f is square integrable and twice differentiable, with all of its second-order partial derivatives bounded, continuous and square integrable.

A2 The kernel K is a square-integrable, spherically symmetric function on $\mathbb{R}^m$ that satisfies $\int K(\mathbf{u}) d\mathbf{u} = 1$, $\int \mathbf{u} K(\mathbf{u}) d\mathbf{u} = \mathbf{0}$ and $\int \mathbf{u} \mathbf{u}^T K(\mathbf{u}) d\mathbf{u} = \mu_2(K) \mathbf{I}_m$ for some $\mu_2(K) > 0$.

A3 The bandwidth matrix $\mathbf{H}$ is a sequence of symmetric positive-definite matrices such that $\mathbf{H} \rightarrow \mathbf{O}$ and $n|\mathbf{H}|^{1/2} \rightarrow \infty$ as $n \rightarrow \infty$.

We consider matrices of the form $\mathbf{H} = ah_n^2 \mathbf{I}_m$ for a fixed constant $a > 0$, where $h_n \in \mathbb{R}$ is a length scale computed from the sample $\{\mathbf{y}_1, \dots, \mathbf{y}_n\}$. Then $|\mathbf{H}|^{1/2} = a^{m/2} h_n^m$, so the two requirements of A3 reduce to $h_n \rightarrow 0$ and $nh_n^m \rightarrow \infty$, which we take as the definition of an admissible bandwidth.

Definition 3 (Admissible bandwidth). We say that h_n is an *admissible bandwidth* for samples in $\mathbb{R}^m$ if $h_n \rightarrow 0$ and $nh_n^m \rightarrow \infty$ as $n \rightarrow \infty$ in probability.

1.4 The lookout algorithm for anomaly detection

The *lookout* algorithm of [Kandanaarachchi & Hyndman \(2022\)](#) has four inputs: the dataset $\{\mathbf{y}_1, \dots, \mathbf{y}_n\}$, where each $\mathbf{y}_i \in \mathbb{R}^m$; a parameter $\beta \in (0, 1)$ indicating the proportion of points to be used for fitting the Generalized Pareto Distribution (GPD); a parameter $\alpha \in (0, 1)$ signifying the cut-off threshold for determining an anomaly; and a logical variable *unitize* indicating whether to scale the data. We briefly outline the steps below.

Algorithm 1 Lookout algorithm

Require: Data $\{\mathbf{y}_1, \dots, \mathbf{y}_n\}$, proportion $\beta \in (0, 1)$, threshold $\alpha \in (0, 1)$, *unitize* $\in \{\text{True}, \text{False}\}$.

- 1: If *unitize*, scale the data to lie in the unit cube in $\mathbb{R}^m$.
 - 2: Compute the persistence homology barcode of the (possibly scaled) data cloud for dimension zero using the Vietoris-Rips diameter ([Ghrist 2008](#)).
 - 3: From the barcode, obtain the ordered finite death diameters $\{d_i\}_{i=1}^{n-1}$ for connected components, and their successive differences $\Delta d_i = d_{i+1} - d_i$.
 - 4: Let $\mathcal{I} = \{i : d_i \geq \text{median}(d_1, \dots, d_{n-1})\}$ and let $d_* = d_{i_*}$ where $i_* = \arg \max\{\Delta d_i : i \in \mathcal{I}\}$; that is, d_* is the diameter corresponding to the lower point of the maximum Δd_i , where the search is restricted to death diameters at or above the median.
 - 5: Compute the kernel density estimates at the observations: $f_i = \hat{f}(\mathbf{y}_i)$, $i = 1, \dots, n$, where $\hat{f}$ is given by (3), $\mathbf{H} = h_n \mathbf{I}_m$, and $h_n = d_*$.
 - 6: Compute the leave-one-out kde values $f_{-i} = \frac{1}{n-1} (nf_i - |\mathbf{H}|^{-1/2} K(\mathbf{0}))$ for $i = 1, \dots, n$.
 - 7: Fit a Generalized Pareto Distribution (GPD) to the largest $1 - \beta$ of the surprisals $\{-\log f_i\}_{i=1}^n$, obtaining estimates of the location, scale and shape parameters (μ, σ, ξ) .
 - 8: Calculate the probability of each observation by applying the estimated GPD to the leave-one-out kde values: $p_i = (1 - \beta)(1 - \hat{F}(-\log f_{-i}))$, where $\hat{F}$ is the distribution function of the fitted GPD.
 - 9: If $p_i < \alpha$, declare $\mathbf{y}_i$ anomalous.
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The lookout algorithm provides density-based anomaly detection, where anomalies are identified as

points with low density in the space of the data. If there are no anomalies in the data, the algorithm should identify α of the data as false positive “anomalies”.

In degree zero the persistent homology of steps 2–4 of [Algorithm 1](#) is a familiar statistical object. Components of the Vietoris-Rips complex merge as the shortest edge between them appears, which is Kruskal’s algorithm, so the merges occur along the edges of the Euclidean minimum spanning tree T of the data. The finite death diameters $d_1 \leq \dots \leq d_{n-1}$ are therefore the ordered edge lengths $e_{(1)} \leq \dots \leq e_{(n-1)}$ of T , and also the merge heights of the single linkage dendrogram ([Gower & Ross 1969](#)). We use the tree rather than the filtration for the rest of the paper, allowing us to draw on the limit theory for minimum spanning trees.

While [Kandanaarachchi & Hyndman \(2022\)](#) demonstrated excellent results in applying their algorithm to a variety of simulated and real datasets, we see several potential areas for improvement that we aim to address in this paper. First, elements of the algorithm are not robust to outliers, including the min-max scaling, and the use of the largest gap in the Rips death diameters (equivalently, edge lengths). Second, the diagonal bandwidth matrix $\mathbf{H}$ does not account for correlations between variables, which can lead to suboptimal performance if the data have strong correlations. Third, it is not obvious that this bandwidth matrix choice even has the right order, or that it shrinks to zero at all, particularly for distributions with heavy tails. Finally, the GPD estimation made minimal assumptions about the distribution of the surprisal values, but we know that these are bounded above and below, and that fact can be used to improve the estimated GPD.

[Section 2](#) defines the bandwidth we use, an upper quantile $e_{(\gamma(n-1))}$ of the edge lengths, and places it among the existing minimum spanning tree and nearest-neighbour methods for anomaly detection. [Section 3](#) then establishes its order. We give matching lower and upper bounds showing that $e_{(\gamma(n-1))}$ is of order $n^{-1/m}$ with high probability, and we exhibit a heavy-tailed distribution for which the previous choice $h_n = d_*$ fails even to converge to zero, so that the change of rule is necessary and not merely convenient.

The mathematical tool behind that heavy-tailed example is the notion of crackle, introduced in [Adler et al. \(2014\)](#), which describes the persistent homology of noise. They proved asymptotic results for the persistent homology of the union of unit balls centered on iid samples of various distributions. There

are qualitative differences between sets of samples taken from the Gaussian distribution (which do not “crackle”) compared to those from heavy-tailed distributions. We adapt their methods to this setting, where we wish to shrink the radius to 0, albeit slowly.

1.5 The modified lookout algorithm

As a result of these considerations, we propose three modifications to the lookout algorithm, described in detail in [Section 4](#).

1. Rather than use a max-min scaling of the data, we standardize the data using a robust covariance estimator. That is, we scale and rotate the data to ensure that the covariance matrix of the transformed data is approximately the identity matrix. This ensures that the data have zero pairwise correlations, which leads to more efficient kernel density estimates. It also leads to each variable having approximately unit variance, which provides a more robust scaling than min-max scaling.
2. Rather than using the lower point of the largest gap in the minimum spanning tree edge lengths, we use an upper quantile of those edge lengths. This avoids the problem where a few anomalies, sitting at the top of the edge length distribution, determine the largest gap and hence the bandwidth. It also gives the bandwidth the correct order, $n^{-1/m}$, under relatively weak conditions.
3. The estimated GPD shape parameter is constrained to be non-positive, which improves the stability of the GPD fitting. This is justified by noting that the kernel density estimates are bounded away from zero, so the surprisals are bounded above and the fitted GPD cannot have a positive shape parameter.

Note that we do not need to estimate the density of the underlying distribution directly. Instead, we can estimate the density of a transformation of the data, and identify anomalies in the transformed space. Hence, there is no need to undo the scaling or rotation.

1.6 Outline of the paper

The remainder of this paper is organized as follows. In [Section 2](#), we define the bandwidth as a quantile of the edge lengths of the Euclidean minimum spanning tree, and relate it to existing minimum spanning

tree and nearest-neighbour methods. In [Section 3](#), we establish the order of that bandwidth and examine what it implies for the kernel density estimates it produces. In [Section 4](#), we propose a modified lookout algorithm that improves performance and robustness under a broader range of conditions. We provide some empirical experiments, demonstrating the performance of the modified lookout algorithm on simulated and real datasets, first in contrast to the original lookout ([Section 5](#)), and then in contrast to other anomaly detection methods ([Section 6](#)). Finally, in [Section 7](#), we summarize our results and discuss future work.

2 Bandwidth selection from the minimum spanning tree

2.1 The bandwidth rule

Let T denote the Euclidean minimum spanning tree of $\{\mathbf{y}_1, \dots, \mathbf{y}_n\} \subset \mathbb{R}^m$: the connected graph on the n observations whose total edge length is smallest. Provided the pairwise distances are distinct, which holds almost surely when the observations are drawn from a density, T is unique and has $n - 1$ edges. Write $e_{(1)} \leq \dots \leq e_{(n-1)}$ for its edge lengths in increasing order.

Definition 4 (Minimum spanning tree bandwidth). For $\gamma \in (0, 1)$, the *minimum spanning tree bandwidth* is the γ quantile of the edge lengths of T ,

$$h_n = e_{(\lceil \gamma(n-1) \rceil)}. \quad (5)$$

We take $\gamma = 0.98$ throughout unless stated otherwise, and in practice we use the median-unbiased quantile estimator of [Hyndman & Fan \(1996\)](#) rather than the order statistic itself; the distinction is immaterial asymptotically. The kernel density estimate [\(3\)](#) is then computed with the bandwidth matrix

$$\mathbf{H} = (m + 4)h_n^2 \mathbf{I}_m, \quad (6)$$

so that h_n is a *length* on the scale of the data rather than a variance. The constant $m + 4$ makes the per-coordinate standard deviation of the spherically symmetric Epanechnikov kernel equal to h_n . In [Section 4](#) the estimate is applied to robustly standardized data, so that a single scalar h_n is enough; the bandwidth matrix in the original coordinates is $(m + 4)h_n^2 \hat{\Sigma}$.

Definition 4 is not how the rule was originally derived. [Kandanaarachchi & Hyndman \(2022\)](#) obtained the bandwidth from the degree-zero persistent homology of a Vietoris-Rips filtration, and [Algorithm 1](#) is stated in those terms. The two descriptions coincide.

Proposition 5. *Let $\{\mathbf{y}_1, \dots, \mathbf{y}_n\} \subset \mathbb{R}^m$ have distinct pairwise distances. The following three sets of $n - 1$ numbers are equal.*

1. *The finite death diameters of the degree-zero persistent homology of the Vietoris-Rips filtration.*
2. *The edge lengths of the Euclidean minimum spanning tree T .*
3. *The merge heights of the single linkage dendrogram.*

Proof. In the Vietoris-Rips complex at diameter t , two observations are joined whenever they lie within distance t of each other. As t increases, edges appear in increasing order of length, and the number of connected components falls by one exactly when an edge appears that joins two distinct components; each such event is the death of a degree-zero homology class, and the death diameter is the length of that edge. Adding edges in increasing order of length and keeping those that join distinct components is Kruskal’s algorithm, so the edges at which deaths occur are precisely the edges of T . That single linkage merges two clusters at the length of the shortest edge between them, and therefore produces the same sequence of merges, is the equivalence of [Gower & Ross \(1969\)](#). $\square$

Remark 6 (Convention). *The scale parameter is a distance between two observations, not the radius of a ball. Two observations lying in a common ball of radius r are therefore joined at diameter $2r$. Working with edge lengths removes the choice: a minimum spanning tree has no radius convention. A reader who prefers radii should halve every scale in this paper, which changes no conclusion and no rate.*

Remark 7 (Persistent homology). *[Proposition 5](#) is the only point at which the filtration is needed. Everything that follows is a statement about the edge lengths of T , and the filtration reappears only in [Section 3.2](#), where the crackle results of [Adler et al. \(2014\)](#) describe the persistent homology of noise.*

The single linkage reading of (5) is what makes the rule interpretable. Cutting the single linkage dendrogram at height h_n leaves the n observations in about $(1 - \gamma)n$ connected components, since each

of the $\lceil \gamma(n-1) \rceil$ shortest edges has already been used to merge two components. With $\gamma = 0.98$, the bandwidth is the scale at which single linkage has gathered all but about 2% of the observations into components. It is a measure of how far apart neighbouring observations are once the few most isolated ones are set aside, which is exactly the scale a density estimate should resolve.

2.2 Relation to existing methods

Minimum spanning trees have a long history in multivariate statistics, and it is worth being explicit about which part of that history we are drawing on.

[Rohlf \(1975\)](#) proposed what is still the standard minimum spanning tree test for multivariate outliers: order the edge lengths, look for a gap near the top, and declare the observations beyond the gap to be outliers. The bandwidth rule of [Kandanaarachchi & Hyndman \(2022\)](#) is a form of the same test. It takes the largest gap among the edge lengths at or above the median and uses the lower endpoint of that gap as the bandwidth. Where Rohlf uses the gap to label points, [Kandanaarachchi & Hyndman \(2022\)](#) use it to set a scale, but both read the same feature of the same distribution, and both inherit the same difficulty: the largest gap near the top of the edge-length distribution is determined by the few most isolated observations, which are the very points an anomaly detector is trying not to be influenced by. [Section 3.2](#) makes this precise, by exhibiting a density for which the top of the edge-length distribution does not settle down at all while a fixed quantile does.

Related uses of the tree include single linkage detectors, which cut the dendrogram and declare the resulting small clusters anomalous, and minimum spanning tree clustering with outlier removal, which deletes long edges before clustering. [Friedman & Rafsky \(1979\)](#) give the broader statistical setting, using minimum spanning trees to construct multivariate analogues of the Wald-Wolfowitz and Smirnov two-sample tests.

Two of the detectors we compare against in [Section 6](#) are built on nearest-neighbour distances combined with extreme value theory: HDoutliers ([Wilkinson 2017](#), [Fraleigh 2018](#)) and stray ([Talagala et al. 2021](#)). The connection to what we do is close. By the cut property of minimum spanning trees, the edge joining an observation to its nearest neighbour is the unique shortest edge across the cut separating that observation from the rest, and so belongs to T ; every nearest-neighbour edge is an edge of the tree. The minimum

spanning tree quantile is thus a connectivity-aware global summary of the same distances, and [Section 3.1](#) shows that it has the same $n^{-1/m}$ order as the typical nearest-neighbour distance.

The distinction that matters is what the tree is used for. We do not use the minimum spanning tree to label anomalies. We use the distribution of its edge lengths only to choose a scale. Labelling is done by the leave-one-out kernel density estimate and the fitted generalized Pareto distribution, exactly as in [Algorithm 1](#). A long edge of T therefore does not make its endpoint an anomaly, and an observation can be anomalous without being incident to a long edge.

2.3 Computation

Computing h_n requires the minimum spanning tree and nothing else. We use the dual-tree Borůvka algorithm of [March et al. \(2010\)](#), as implemented in `mlpack` ([Singh Parihar et al. 2026](#)), which builds the exact Euclidean minimum spanning tree in $O(n \log n)$ time under the same assumptions that make dual-tree nearest-neighbour search efficient, and degrades gracefully in higher dimensions. The naive alternative costs $O(n^2)$ in time and memory, since it forms all pairwise distances.

This is a change of implementation as well as of description. A Vietoris-Rips filtration builds simplices in every degree and is far more expensive than the degree-zero information we actually use; computing it to extract $n - 1$ numbers that a minimum spanning tree algorithm returns directly is wasteful. The `lookout` package computes the tree.

The tree is built on all n observations, so the bandwidth of [Definition 4](#) is exactly the quantile whose behaviour [Section 3](#) describes, with no subsampling or other approximation in between.

3 Asymptotics of the minimum spanning tree bandwidth

This section establishes the order of $h_n = e_{(\lceil \gamma(n-1) \rceil)}$, shows why a fixed quantile is the right thing to take and the largest gap is not, and works out what the resulting bandwidth does to the kernel density estimate. Throughout, b_m denotes the volume of the unit ball in $\mathbb{R}^m$, M a global upper bound on the density f , and $\{\omega_n\}$ a sequence of integers indexing which edge length is used. Proofs of [Theorem 8](#) and of [Counterexample 14](#) are in [Appendix A](#).

3.1 The order of the quantile edge length

The upper bound holds under a local Lipschitz condition and no assumption at all on the tails.

Theorem 8. *Let $e_{(k)}$ denote the k th smallest edge length of the Euclidean minimum spanning tree of n points sampled independently from a distribution on $\mathbb{R}^m$ whose density f is Lipschitz with constant M over the region where $f \leq 1$. Fix $\gamma \in (0, 1)$ and choose any sequence of integers $\{\omega_n\}$ with $0 \leq \omega_n \leq \gamma n$ for all sufficiently large n . Then there exists a constant $C(f, \gamma)$, depending on f and γ but not on n , such that*

$$\mathbb{P}(e_{(\omega_n)} \leq C(f, \gamma)n^{-1/m}) \geq 1 - \frac{12}{n(1 - \gamma)}$$

for all sufficiently large n .

The argument is a covering one. Choose a superlevel set $L = f^{-1}[c, \infty)$ carrying all but a small fraction of the probability, cover it by $O(r_n^{-m})$ balls of radius r_n , and note that two points in a common ball are joined at diameter $2r_n$, so the number of components at that diameter is at most the number of balls plus the number of sample points outside L . Choosing $r_n \asymp n^{-1/m}$ makes the first count at most $n(1 - \gamma)/2$, and Chebyshev's inequality controls the second. The constant is $C(f, \gamma) = 2(2^{m+1}/(cb_m(1 - \gamma)))^{1/m}$.

A matching lower bound holds under the weaker assumption that f is merely bounded.

Theorem 9. *Let $e_{(k)}$ be as in [Theorem 8](#), with f bounded above by M . Fix $\gamma \in (0, 1)$ and choose any sequence of integers $\{\omega_n\}$ with $\gamma n \leq \omega_n < n$ for all sufficiently large n . Put $c = (\gamma/(Mb_m))^{1/m}$. Then, for all sufficiently large n ,*

$$\mathbb{P}(e_{(\omega_n)} \geq cn^{-1/m}) \geq 1 - \frac{6}{\gamma n}.$$

Proof. Let $E(n, r) = \sum_{i < j} \mathbf{1}\{\|\mathbf{y}_i - \mathbf{y}_j\| \leq r\}$ be the number of pairs within distance r , and let $C(n, r)$ be the number of connected components of the graph joining all such pairs. The number of minimum spanning tree edges of length at most r is $n - C(n, r)$, and adding an edge either reduces the component count by one or leaves it unchanged, so $n - C(n, r) \leq E(n, r)$. Hence

$$\{e_{(\omega_n)} < r\} \subseteq \{n - C(n, r) \geq \omega_n\} \subseteq \{E(n, r) \geq \gamma n\}.$$

Set $r_n = cn^{-1/m}$, so that $\theta := Mb_m r_n^m = \gamma/n$. Since $\mathbb{P}(\|\mathbf{y}_1 - \mathbf{y}_2\| \leq r) \leq Mb_m r^m$,

$$\mathbb{E}[E(n, r_n)] = \binom{n}{2} \mathbb{P}(\|\mathbf{y}_1 - \mathbf{y}_2\| \leq r_n) \leq \frac{n(n-1)}{2} \cdot \frac{\gamma}{n} < \frac{\gamma n}{2}.$$

Now $E(n, r_n)$ is an unnormalised U-statistic of degree two with kernel $h_{ij} = \mathbf{1}\{\|\mathbf{y}_i - \mathbf{y}_j\| \leq r_n\}$, so

$$\text{Var}(E(n, r_n)) = \binom{n}{2} \text{Var}(h_{12}) + n(n-1)(n-2) \text{Cov}(h_{12}, h_{13}).$$

The first term is at most $\binom{n}{2}\theta < \gamma n/2$. For the second, conditioning on $\mathbf{y}_1$ gives $\text{Cov}(h_{12}, h_{13}) \leq \mathbb{E}[\mathbb{P}(\mathbf{y}_2 \in B(\mathbf{y}_1, r_n) \mid \mathbf{y}_1)^2] \leq \theta^2$, so that term is at most $n^3\theta^2 = \gamma^2 n$. Therefore $\text{Var}(E(n, r_n)) \leq \frac{3}{2}\gamma n$. Since $\mathbb{E}[E(n, r_n)] < \gamma n/2$, Chebyshev's inequality gives

$$\mathbb{P}(E(n, r_n) \geq \gamma n) \leq \mathbb{P}\left(E(n, r_n) - \mathbb{E}[E(n, r_n)] > \frac{\gamma n}{2}\right) \leq \frac{\frac{3}{2}\gamma n}{(\gamma n/2)^2} = \frac{6}{\gamma n}. \quad \square$$

Markov's inequality alone is not enough here: it bounds $\mathbb{P}(E(n, r_n) \geq \gamma n)$ by a constant below one, which does not vanish. The variance calculation is what turns the bound into a statement that holds with probability tending to one, and it is why earlier versions of this result had to be weakened by a factor tending to zero.

Together the two theorems pin down the order exactly.

Corollary 10. *Under the assumptions of [Theorem 8](#), there exist constants $0 < c \leq C < \infty$ depending on f and γ such that*

$$\mathbb{P}(c n^{-1/m} \leq e_{(\lceil \gamma(n-1) \rceil)} \leq C n^{-1/m}) \rightarrow 1 \quad \text{as } n \rightarrow \infty;$$

that is, $h_n \asymp n^{-1/m}$ in probability.

Note that both bounds fail at γ close to its endpoints, and in opposite ways: the upper bound degrades as $\gamma \rightarrow 1$ and the lower bound as $\gamma \rightarrow 0$. At the default $\gamma = 0.98$ the guaranteed constants are one to two orders of magnitude away from the observed values, so [Corollary 10](#) should be read as a statement about order and not as a usable numerical bound.

Remark 11 (Total tree length). *For light-tailed densities there is a sharper route to the upper bound. [Steele \(1988\)](#) showed for compactly supported densities, and [Penrose & Yukich \(2003, Theorem 1.1\)](#) for densities on $\mathbb{R}^m$ satisfying a moment condition, that the total length M_n of the minimum spanning tree satisfies $n^{1/m-1}M_n \rightarrow C(1, m) \int f^{1-1/m}$ almost surely; the proof is given in [Yukich \(1998\)](#). If $e_{(k)} > t$ then all $n - k$ edges ranked k and above exceed t , so $M_n \geq (1 - \gamma)(n - 1)t$, and therefore*

$$h_n \leq \frac{M_n}{(1 - \gamma)(n - 1)} = \frac{C(1, m) \int f^{1-1/m}}{1 - \gamma} n^{-1/m} (1 + o(1))$$

almost surely. The constant is explicit and interpretable, where the constant of [Theorem 8](#) involves a covering number. This complements [Theorem 8](#) rather than replacing it: the moment condition fails for the heavy-tailed density of [Section 3.2](#), which is precisely the case the paper needs, and there the covering argument, which requires no moment at all, is the only one of the two that survives.

Remark 12 (Nearest-neighbour distances). *It is natural to ask why the tree is needed when nearest-neighbour distances have the same $n^{-1/m}$ scaling and are cheaper still. The two are not alternatives: by the cut property, the edge from an observation to its nearest neighbour is the unique shortest edge across the cut isolating that observation, so it belongs to the minimum spanning tree. Every nearest-neighbour distance is therefore an edge length of T , and [Corollary 10](#) says the γ quantile of all $n - 1$ edge lengths has the same order as the typical nearest-neighbour distance. What the tree adds is the remaining edges, which record how the components of the data join up rather than only how close each point is to its closest companion. [Section 6](#) compares lookout against detectors built directly on nearest-neighbour distances.*

Remark 13 (The longest edge). *The results above say nothing about $e_{(n-1)}$, the longest edge, which is the object a max-gap rule is closest to. Its behaviour is known in considerable detail: [Penrose \(1997\)](#) gives the limiting distribution of the longest edge of the minimum spanning tree for densities on a compact region, and [Penrose \(1998\)](#) treats the normally distributed case. Both are sharper than anything we prove or need, and we refer the reader to them rather than restating weaker versions here.*

3.2 Why a fixed quantile and not the largest gap

The original bandwidth rule takes the largest gap in the upper half of the edge lengths, which places it at the extreme top of the edge-length distribution, where $\omega_n/n \rightarrow 1$. [Corollary 10](#) does not apply there, and for good reason: at heavy tails the top of the distribution need not settle down at all.

Counterexample 14. *Let $m \geq 3$ and let the observations be sampled from the density $f(\mathbf{x}) = c_m/(1 + \|\mathbf{x}\|^{m+1})$ on $\mathbb{R}^m$. For $\omega_n = n - \frac{1}{4}n^{1-2/m}$ we have $\mathbb{P}(e_{(\omega_n)} > 1) > 1/4$ for all sufficiently large n , so $e_{(\omega_n)}$ does not converge to zero.*

The mechanism is crackle ([Adler et al. 2014](#)). Under a heavy-tailed density the sample has, with probability bounded away from zero, a halo of order $n^{1-2/m}$ points at radius beyond $R_n \asymp n^{2/m}$ that are

mutually isolated at unit scale, each forming its own connected component. Any rule that reads the top $O(n^{1-2/m})$ of the edge-length distribution is reading that halo, and the halo does not shrink. A fixed quantile $\gamma < 1$ never looks that far up, and [Theorem 8](#) shows it converges at the right rate for this density as for any other. The full argument is in [Appendix A](#).

There is a second way to see where the difficulty lives. The integral $\int f^{1-1/m}$ that governs the total tree length in [Remark 11](#) is finite for compactly supported and Gaussian densities, but for the density above $f^{1-1/m}$ decays like $\|\mathbf{x}\|^{-(m^2-1)/m}$ and $(m^2-1)/m < m$, so the integral diverges and the total tree length grows faster than $n^{1-1/m}$. The excess is absorbed entirely by the longest edges: the γ quantile still behaves like $n^{-1/m}$ while the mean edge length does not. The same integral that controls the scaling is the one that fails under heavy tails, and it fails exactly where the max-gap rule looks and the fixed quantile does not.

This is the formal version of the objection raised in [Section 2.2](#). [Rohlf \(1975\)](#) reads a gap near the top of the edge lengths to find outliers, and reading the same feature to set a bandwidth inherits its instability.

3.3 The bandwidth sits at the critical scaling

We can now say what [\(6\)](#) does to the kernel density estimate.

Proposition 15. *Let f satisfy the assumptions of [Theorem 8](#) and let $\mathbf{H} = (m+4)h_n^2\mathbf{I}_m$ with $h_n = e_{(\lceil \gamma(n-1) \rceil)}$. Then there exist constants $0 < c' \leq C' < \infty$ such that $\mathbb{P}(c' \leq n|\mathbf{H}|^{1/2} \leq C') \rightarrow 1$ as $n \rightarrow \infty$.*

Proof. $|\mathbf{H}|^{1/2} = (m+4)^{m/2}h_n^m$, so $n|\mathbf{H}|^{1/2} = (m+4)^{m/2}nh_n^m$, and [Corollary 10](#) gives nh_n^m between c^m and C^m with probability tending to one. $\square$

Condition A3 therefore fails: $n|\mathbf{H}|^{1/2}$ is bounded rather than divergent, the variance term $n^{-1}|\mathbf{H}|^{-1/2}R(K)$ of the MISE does not vanish, and the estimator is not MISE-consistent. This is a property of the rule, not an oversight, and it has a clean reading. The expected number of observations falling inside the kernel window at a typical point is of order $n|\mathbf{H}|^{1/2}f$, which [Proposition 15](#) says is $O(1)$ in n : the estimator has a fixed local resolution, set by a fixed number of neighbours rather than by a shrinking neighbourhood. That is the same design as a fixed- k nearest-neighbour density estimate,

which is likewise inconsistent and which underlies both the local outlier factor (Breunig et al. 2000) and nearest-neighbour outlier scores. Section 3.4 argues that this is the right place to be.

Remark 16 (Consistency is one factor away). *Nothing prevents consistency if it is wanted. For any $a_n \rightarrow \infty$, however slowly, the inflated bandwidth $a_n h_n$ satisfies $a_n h_n \rightarrow 0$ and $n(a_n h_n)^m \asymp a_n^m \rightarrow \infty$, so it is an admissible bandwidth in the sense of Definition 3 and the resulting estimator is MISE-consistent. Taking $a_n = \log n$ changes the bandwidth by a factor invisible at the sample sizes considered here. The rule sits at the boundary by choice.*

3.4 Why MISE-optimality is the wrong target

An asymptotic approximation to the MISE (4) is given by Chacón & Duong (2018):

$$\text{AMISE}(\hat{f}) = n^{-1} |\mathbf{H}|^{-1/2} R(K) + \frac{1}{4} \mu_2(K)^2 \int \{\text{tr}(\mathbf{H} D^2 f(\mathbf{u}))\}^2 d\mathbf{u}, \quad (7)$$

where $R(K) = \int K(\mathbf{u})^2 d\mathbf{u}$ and $D^2 f$ is the Hessian of f . Writing $\mathbf{H} = h^2 \mathbf{I}_m$ with h a length, this is

$$\text{AMISE} = n^{-1} h^{-m} R(K) + \frac{1}{4} \mu_2(K)^2 h^4 \int \{\Delta f(\mathbf{u})\}^2 d\mathbf{u}, \quad (8)$$

minimised at $h_{\text{opt}} \asymp n^{-1/(m+4)}$, with MISE of order $n^{-4/(m+4)}$. Against this, Corollary 10 gives $h_n \asymp n^{-1/m}$. The ratio is

$$\frac{h_n}{h_{\text{opt}}} \asymp n^{-4/\{m(m+4)\}},$$

which tends to zero for every $m \geq 1$. The minimum spanning tree bandwidth undersmooths relative to MISE in every dimension, severely when m is small and mildly when m is large. At $h \asymp n^{-1/m}$ the bias term of (8) is $O(n^{-4/m})$ and vanishes, while the variance term is $\Theta(1)$; all of the residual error is variance.

We argue that this is acceptable, and for anomaly detection desirable. The MISE is an integrated criterion weighted by the density itself, so it is dominated by the bulk of the distribution, and the bandwidth that minimises it is the one that best describes where most of the data are. Anomaly detection asks a different question. What matters is the ranking of $\hat{f}$ at the observed points in the low-density region, and the upper tail of the surprisals $-\log \hat{f}(\mathbf{y}_i)$, neither of which is what (7) measures. A bandwidth chosen for the bulk oversmooths the tail, filling in the gaps around isolated observations with probability mass borrowed from their distant neighbours, which is precisely the signal an anomaly detector needs to keep. Undersmoothing keeps isolated points isolated.

The guarantee to demand is therefore not consistency and not an optimal rate, but the correct order: h_n should be the scale at which neighbouring observations sit, which is $n^{-1/m}$. That is what [Theorem 8](#) and [Theorem 9](#) deliver, and it is what rules out the largest-gap rule, whose bandwidth fails even to converge to zero under the heavy tails of [Counterexample 14](#). Correct order is a weaker requirement than MISE-optimality, and it is the one that discriminates between the two rules.

4 Lookout algorithm v2

In light of the preceding theoretical considerations, we propose three modifications to the lookout algorithm, described in this section.

4.1 Standardization

The *minimum covariance determinant* estimator provides a robust estimate of Σ , the covariance matrix of $\{\mathbf{y}_1, \dots, \mathbf{y}_n\}$, by searching for the subsample of a given size whose sample covariance matrix has the smallest determinant ([Rousseeuw & Driessen 1999](#)). We take that subsample to be of size $0.9n$. This is less resistant than the usual choice of $n/2$, but anomalies are rare by assumption, and the larger subsample gives a more efficient estimate of Σ for the bulk of the data, which is what the scaling is for. Let

$$\hat{\Sigma}^{-1} = \mathbf{U}'\mathbf{U} \tag{9}$$

denote the Cholesky decomposition of its inverse. Then, the data can be scaled by $\mathbf{U}$, giving $\mathbf{z}_i = \mathbf{U}(\mathbf{y}_i - \mathbf{m})$, where $\mathbf{m}$ is the vector of medians of $\{\mathbf{y}_1, \dots, \mathbf{y}_n\}$. The covariance matrix of the scaled data $\{\mathbf{z}_1, \dots, \mathbf{z}_n\}$ is (approximately) the identity matrix. This can be thought of as both scaling and rotating the data; it is the multivariate equivalent of computing robust z-scores.

We prefer this to the min-max scaling used in the original lookout algorithm, as it is robust to outliers and removes the correlations between variables, leading to more efficient kernel density estimates.

4.2 Bandwidth matrix selection

Let $e_{(1)} \leq \dots \leq e_{(n-1)}$ denote the ordered edge lengths of the Euclidean minimum spanning tree of the scaled and rotated data $\mathbf{z}_1, \dots, \mathbf{z}_n$, and let $h_n = e_{(\lceil \gamma(n-1) \rceil)}$ be the minimum spanning tree bandwidth of

Definition 4, for a large value of $\gamma \in (0, 1)$.

A kernel density estimator is applied to the scaled data $\{\mathbf{z}_1, \dots, \mathbf{z}_n\}$ using the bandwidth matrix $\mathbf{H} = (m + 4)h_n^2 \mathbf{I}_m$, where m is the dimension of the data. By [Proposition 15](#) this places the estimator at the critical scaling, with a fixed number of observations inside the kernel window; [Section 3.4](#) explains why that is the right choice for anomaly detection. When the proportion of anomalies in the data is small, we would expect this estimate to be close to the density of the non-anomalous observations.

Further, the kernels are now aligned with the data, as both have covariance matrices that are scalar multiples of the identity matrix. While alignment is not necessary, it does improve the efficiency of the kernel density estimates ([Chacón & Duong 2018](#)).

An equivalent density estimate of the original data $\{\mathbf{y}_1, \dots, \mathbf{y}_n\}$ could be obtained using $\mathbf{H} = (m + 4)h_n^2 \hat{\Sigma}$, although for our purposes, working with the scaled data is sufficient.

4.3 Bounds for kernel density estimates

In the following lemma we show that the kernel density estimates at the observations, and their negative logarithms, are bounded by constants that depend only on the dimension m and the sample size n .

Lemma 17. *Let $\{\mathbf{y}_1, \dots, \mathbf{y}_n\}$ be n data points in $\mathbb{R}^m$, with kernel density estimate given by*

$$\hat{f}(\mathbf{y}) = \frac{1}{n} \sum_{i=1}^n |\mathbf{H}|^{-1/2} K\left(\mathbf{H}^{-1/2}(\mathbf{y} - \mathbf{y}_i)\right),$$

where $K(\mathbf{u})$ is a multivariate kernel function that is bounded below by 0 and above by $K(\mathbf{0}) = K_0$, and $\mathbf{H}$ is an $m \times m$ symmetric positive-definite bandwidth matrix. Then for all $j \in \{1, \dots, n\}$, the kernel density estimates at the observations, $\hat{f}(\mathbf{y}_j)$, and their negative logarithms, $-\log \hat{f}(\mathbf{y}_j)$, are bounded by constants that depend only on m and n .

Proof.

$$\frac{K_0 |\mathbf{H}|^{-1/2}}{n} \leq \hat{f}(\mathbf{y}_j) = \frac{1}{n} \sum_{i=1}^n |\mathbf{H}|^{-1/2} K\left(\mathbf{H}^{-1/2}(\mathbf{y}_j - \mathbf{y}_i)\right) \leq K_0 |\mathbf{H}|^{-1/2}.$$

The constant K_0 may depend on m . □

For example, if we use the multivariate Gaussian kernel $K(\mathbf{u}) = \frac{1}{(2\pi)^{m/2}} \exp\left(-\frac{1}{2}\|\mathbf{u}\|^2\right)$, then $K_0 = \frac{1}{(2\pi)^{m/2}}$. If we use the multivariate Epanechnikov kernel $K(\mathbf{u}) = \frac{m+2}{2b_m} (1 - \|\mathbf{u}\|^2)_+$, where b_m is the volume of the unit ball in $\mathbb{R}^m$, and $u_+ = \max(0, u)$, then $K_0 = (m + 2)/(2b_m)$.

In particular the surprisals $-\log \hat{f}(\mathbf{y}_j)$ are bounded above by $\log(n/K_0) + \frac{1}{2} \log |\mathbf{H}|$. A generalized Pareto distribution with non-negative shape parameter has unbounded support, so when we fit a generalized Pareto distribution (GPD) to the largest $1 - \beta$ of $\{-\log \hat{f}(\mathbf{y}_j)\}_{j=1}^n$ we may constrain its shape parameter to be non-positive, which improves the stability of the GPD fitting.

Two approximations are involved here. These surprisals are computed from $\hat{f}$ rather than from f , and they are evaluated at the observations rather than at independent draws, so the values to which the GPD is fitted are neither mutually independent nor distributed exactly as the S of [Section 1](#). The peaks-over-threshold model is used as an approximation on both counts.

4.4 Modified lookout algorithm

We are now in a position to describe the modified lookout algorithm.

Algorithm 2 Modified lookout algorithm

Require: Data $\{\mathbf{y}_1, \dots, \mathbf{y}_n\}$, proportion $\beta \in (0, 1)$, threshold $\alpha \in (0, 1)$, and quantile level $\gamma \in (0, 1)$.

- 1: Optional scaling: calculate the Cholesky decomposition (9), and centre, rotate and scale the data using $\mathbf{U}$, giving $\mathbf{z}_i = \mathbf{U}(\mathbf{y}_i - \mathbf{m})$ for $i = 1, \dots, n$, where $\mathbf{m}$ is the vector of medians of $\{\mathbf{y}_1, \dots, \mathbf{y}_n\}$.
 - 2: Compute the Euclidean minimum spanning tree of $\{\mathbf{z}_1, \dots, \mathbf{z}_n\}$ (set $\mathbf{z}_i = \mathbf{y}_i$ if no scaling is applied), and obtain its ordered edge lengths $e_{(1)} \leq \dots \leq e_{(n-1)}$.
 - 3: Let $h_n = e_{(\lceil \gamma(n-1) \rceil)}$, the γ sample quantile computed from $\{e_{(i)}\}_{i=1}^{n-1}$.
 - 4: Compute the kernel density estimates at the scaled observations: $f_i = \hat{f}(\mathbf{z}_i)$, $i = 1, \dots, n$, where $\hat{f}$ is given by (3) applied to $\mathbf{z}_1, \dots, \mathbf{z}_n$ with $\mathbf{H} = (m+4)h_n^2 \mathbf{I}_m$.
 - 5: Compute the leave-one-out kde values $f_{-i} = \frac{1}{n-1} (nf_i - |\mathbf{H}|^{-1/2} K(\mathbf{0}))$ for $i = 1, \dots, n$.
 - 6: Fit a Generalized Pareto Distribution (GPD) to the largest $1 - \beta$ of the surprisals $\{-\log f_i\}_{i=1}^n$, constraining the shape parameter to be non-positive, obtaining estimates of the location, scale and shape parameters (μ, σ, ξ) .
 - 7: Calculate the probability of each observation by applying the estimated GPD to the leave-one-out kde values: $p_i = (1 - \beta)(1 - \hat{F}(-\log f_{-i}))$, where $\hat{F}$ is the distribution function of the fitted GPD.
 - 8: If $p_i < \alpha$, declare $\mathbf{y}_i$ anomalous.
-

As with the original lookout algorithm, the values of p_i can also act as anomaly scores, with lower

values indicating more anomalous points.

The default values for the parameters are $\alpha = 0.001$, $\beta = 0.90$ and $\gamma = 0.98$. The default values are used in all experiments reported here, unless explicitly stated otherwise. Scaling is also applied by default, unless otherwise stated.

Step 4 evaluates (3) at each of the n observations, which costs $O(n^2)$ if the sum is taken over all pairs. For very large n the implementation instead sums over the nearest few hundred neighbours of each observation. Because the Epanechnikov kernel has compact support, this changes nothing whenever fewer than that many observations lie within the kernel’s half-width, which by [Proposition 15](#) is the typical case: the number of observations inside the window does not grow with n .

5 Comparing new lookout with the older version

Next we compare the old and the new versions of lookout. We conduct four experiments that explore different aspects of the updated algorithm, and illustrate the output using a set of showcase examples. All experiments have anomalies and non-anomalies. Experiments 1 and 2 allow the anomalies to move away from the non-anomalous distribution. These two experiments compare the sensitivity of the old and new versions of lookout to identifying anomalies as they become more obvious. Experiments 3 and 4 focus on having anomalies on the boundary of the non-anomalous region. For these two experiments we increase the number of data points and explore the performance of the two algorithm versions for different sample sizes.

5.1 Experiment 1: Different anomaly rates for Gamma distribution

For experiment 1 we consider the $\text{Gamma}(a, r)$ distribution with shape a and rate r and generate data points with different rates r . We select $\text{Gamma}(2, 2)$ as the non-anomalous distribution and $\text{Gamma}(2, r)$ as the anomalous distribution where $r \in \{0.1, 0.2, \dots, 1\}$. Let $Y_i = (Y_{i1}, Y_{i2})$ where Y_{i1} and Y_{i2} are independent with $Y_{ij} \sim \text{Gamma}(2, 2)$ for the non-anomalous distribution and $Y_{ij} \sim \text{Gamma}(2, r)$ for the anomalous distributions. We consider 10 replications of the experiment for each value of r . For each replication we generate 500 non-anomalies and 10 anomalies.

[Fig. 1](#) shows the data and results of Experiment 1. As the rate r of the anomalous Gamma distribution

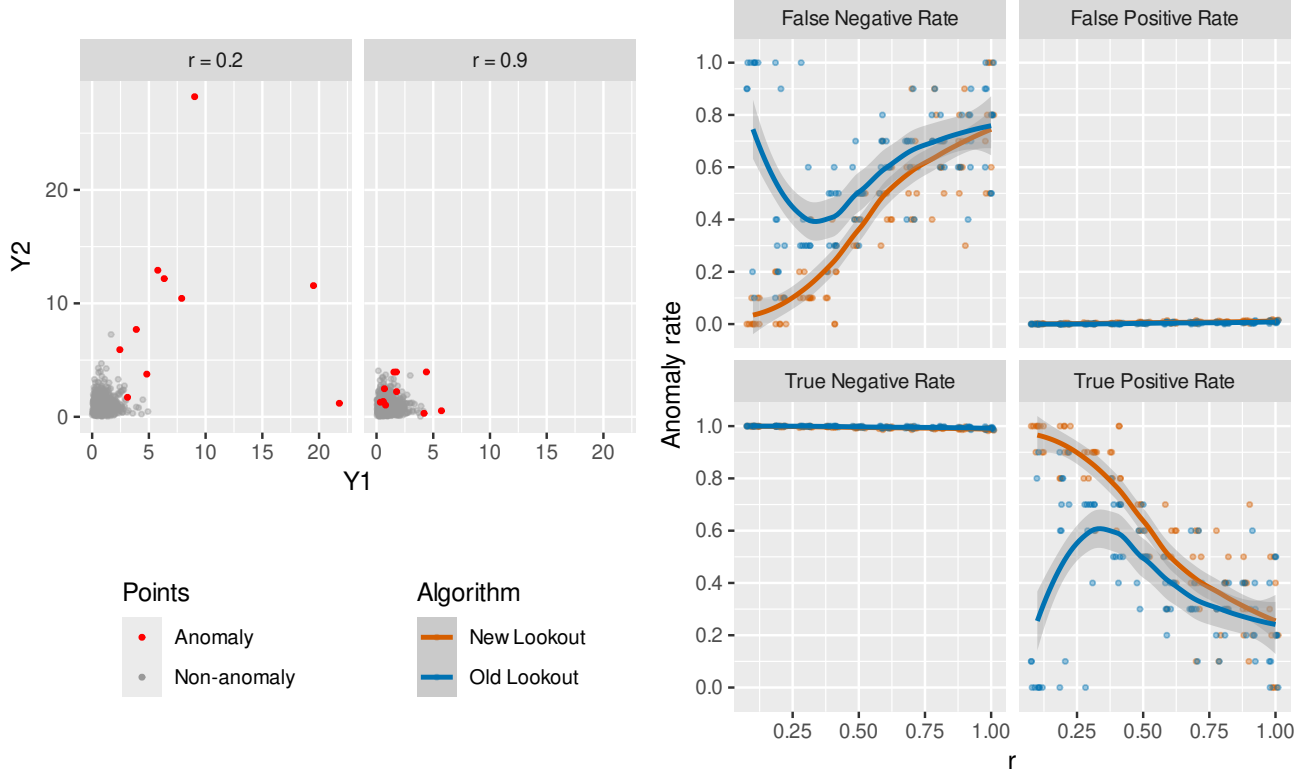


Figure 1: Experiment 1 data and results. Left: data for two iterations of the experiment with $r = 0.2$ and $r = 0.9$. Anomalies with lower values of r are easier to identify. Right: the relative performance of the two algorithms. The points in the right panel are slightly jittered horizontally to avoid overlapping points.

increases, the anomalies and non-anomalies become more similar. Thus, we expect a decrease in the true positive rate and an increase in the false negative rate as r increases. The anomalies are maximally different from non-anomalies when $r = 0.1$. We see that new lookout performs better on average than the old lookout algorithm across all values of r , with the largest differences for the smallest values of r .

5.2 Experiment 2: Different anomaly means for Normal distribution

For experiment 2 we consider the normal distribution $\mathcal{N}(\mu, 1)$ with mean μ . We let $Y_i = (Y_{i1}, Y_{i2})$ where Y_{i1} and Y_{i2} are independent with $Y_{ij} \sim \mathcal{N}(\mu, 1)$. The non-anomalous distribution has $\mu = 0$, while the anomalous distribution has $\mu \in \{2.50, 2.75, \dots, 3.75, 4.00\}$. We consider 1000 non-anomalies and 10 anomalies, and we produce 10 replicates for each value of μ .

Fig. 2 shows Experiment 2 data and results. As μ increases, the anomalies shift away from the non-anomalies. Thus, we expect the true positive rate to increase and the false negative rate to decrease with μ . Here, the two versions of the lookout algorithm are statistically indistinguishable, across all values of μ .

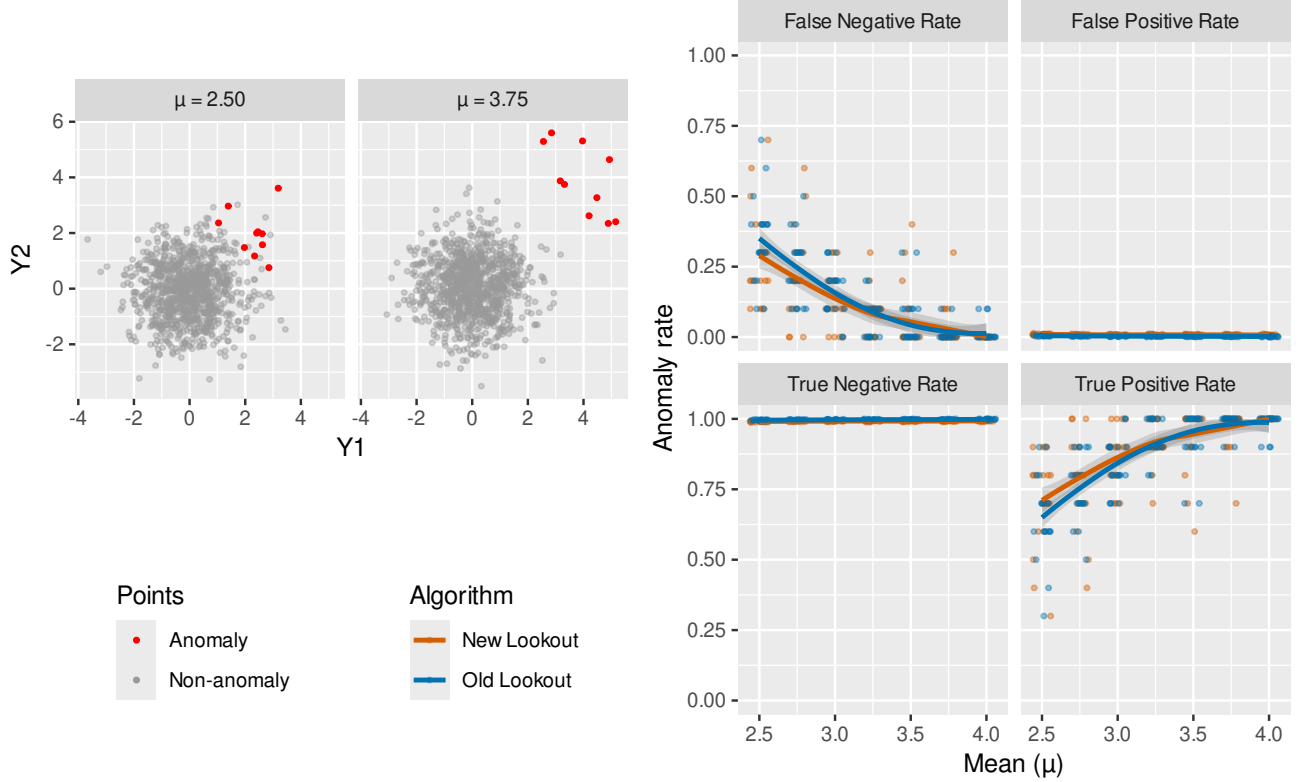


Figure 2: Experiment 2 data and results. Left: data for two iterations of the experiment with $\mu = 2.5$ and $\mu = 3.75$. Anomalies with higher values of μ are easier to identify. Right: the relative performance of the two algorithms. The points in the right panel are slightly jittered horizontally to avoid overlapping points.

5.3 Experiment 3: As N increases - Normal distribution

It is often difficult to distinguish anomalies from legitimate observations because they can appear similar. In such cases, anomalies are not well separated from the non-anomalous points. In this experiment, we explore the efficacy of *lookout* when anomalies are close to regions containing non-anomalous points.

We consider n non-anomalies for $n \in \{1000, 2000, \dots, 10000\}$ and $0.005 \times n$ anomalies. We let $Y_i =$

(Y_{i1}, Y_{i2}) where Y_{i1} and Y_{i2} are independent, with $Y_{ij} \sim \mathcal{N}(\mu, \sigma)$. For the non-anomalies, we set $\mu = 0$ and $\sigma = 1$. We generate anomalies in a ring just beyond most of the non-anomalous points, by first sampling $\mu_{i1} \sim \mathcal{U}(-\sqrt{2} \times 2.2, \sqrt{2} \times 2.2)$ and letting $\mu_{i2} = \sqrt{2 \times 2.2^2 - \mu_{i1}^2}$. Then we sample the anomalies (Y_{i1}, Y_{i2}) where $Y_{i1} \sim \mathcal{N}(\mu_{i1}, 0.1)$ and $Y_{i2} \sim \mathcal{N}(\mu_{i2}, 0.1)$. We produce 10 replications for each value of n .

Fig. 3 shows the data of a single run and the results of experiment 3. Both new and old lookout give low false positive rates and high true negative rates. However, we see that new lookout has a larger true positive rate and a lower false negative rate compared to old lookout, with statistically significant differences for smaller sample sizes.

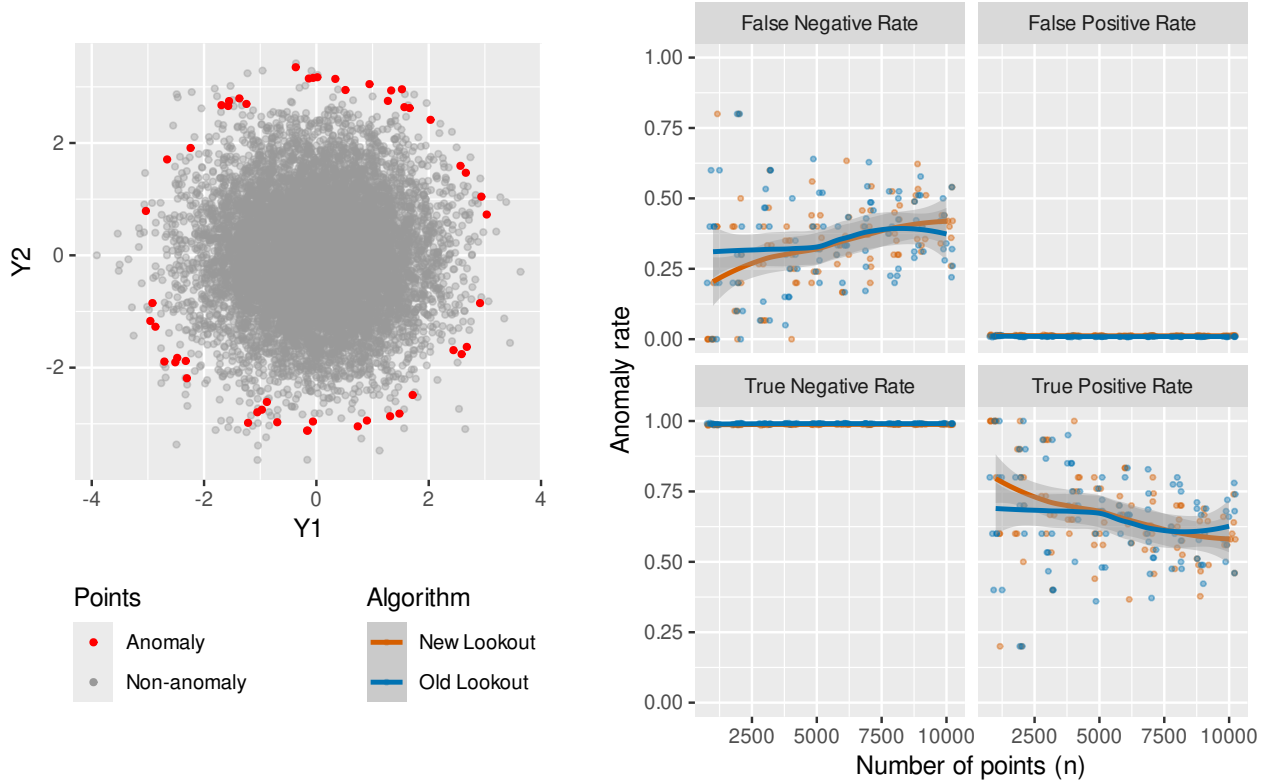


Figure 3: Experiment 3 data and results using Normal distributions. Left: data for $n = 10000$. Right: the relative performance of the two algorithms. The points in the right panel are slightly jittered horizontally to avoid overlapping points.

5.4 Experiment 4: As N increases - Gamma distribution

This experiment is similar to experiment 3, but we set $Y_{ij} \sim \text{Gamma}(2, 2)$ for the non-anomalous points. To obtain anomalous points, we first generate n points $Y_i = (Y_{i1}, Y_{i2})$ where Y_{i1} and Y_{i2} are independent, with $Y_{ij} \sim \text{Gamma}(2.2, 2)$. Then we consider the subset $\{Y_i : \|Y_i\| > \|X\|_{0.99}\}$ where $\|Y_i\| = \sqrt{Y_{i1}^2 + Y_{i2}^2}$ denotes the distance from the origin to Y_i and $\|X\|_{0.99}$ denotes the 0.99 quantile of $\left\{ \sqrt{Y_{i1}^2 + Y_{i2}^2} \right\}_{i=1}^n$. From this subset we sample $0.005 \times n$ points and label them as anomalies. Each iteration for a given n is repeated 10 times. Fig. 4 shows the data from an iteration of the experiment with $n = 10000$ and the results for the new and old version of *lookout*. The two versions of *lookout* are not significantly different, other than at the largest sample sizes, where the older version does slightly better.

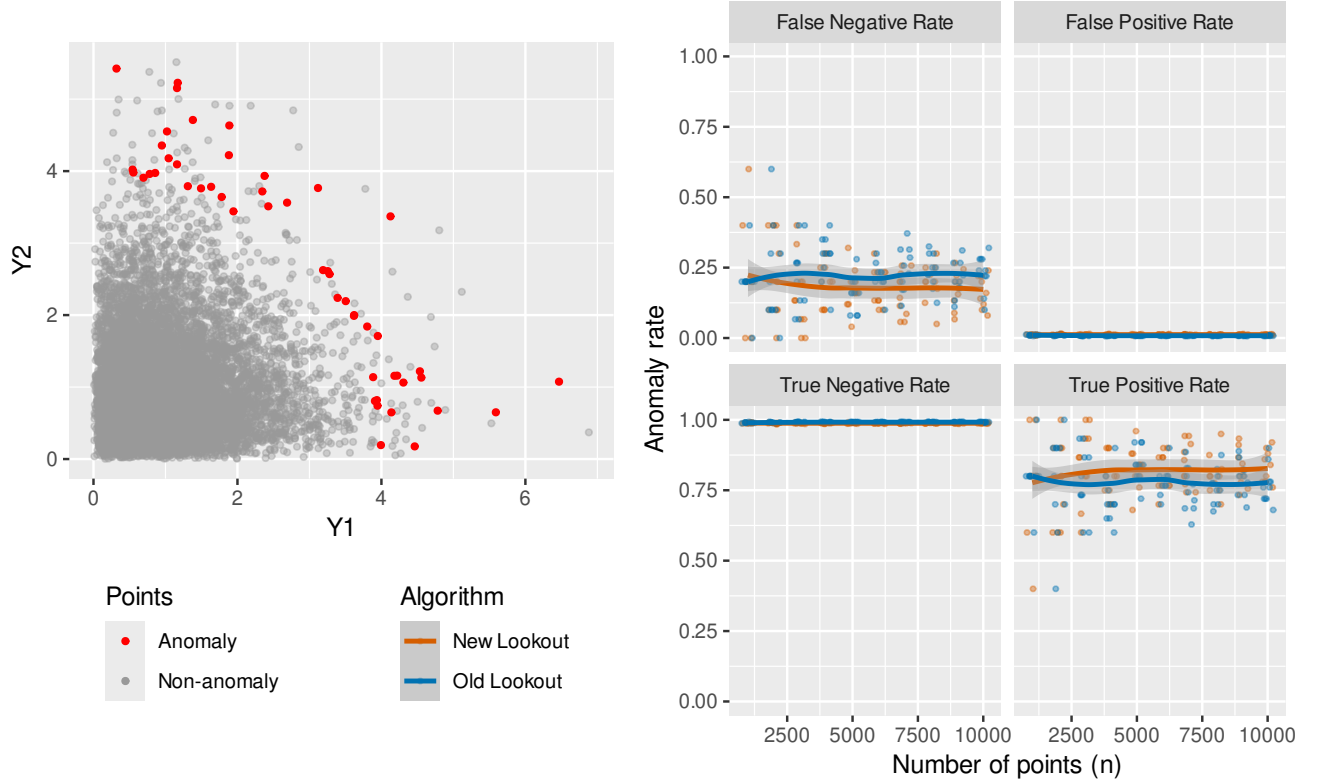


Figure 4: Experiment 4 data and results using Gamma distributions. Left: data for $n = 10000$. Right: the relative performance of the two algorithms. The points in the right panel are slightly jittered horizontally to avoid overlapping points.

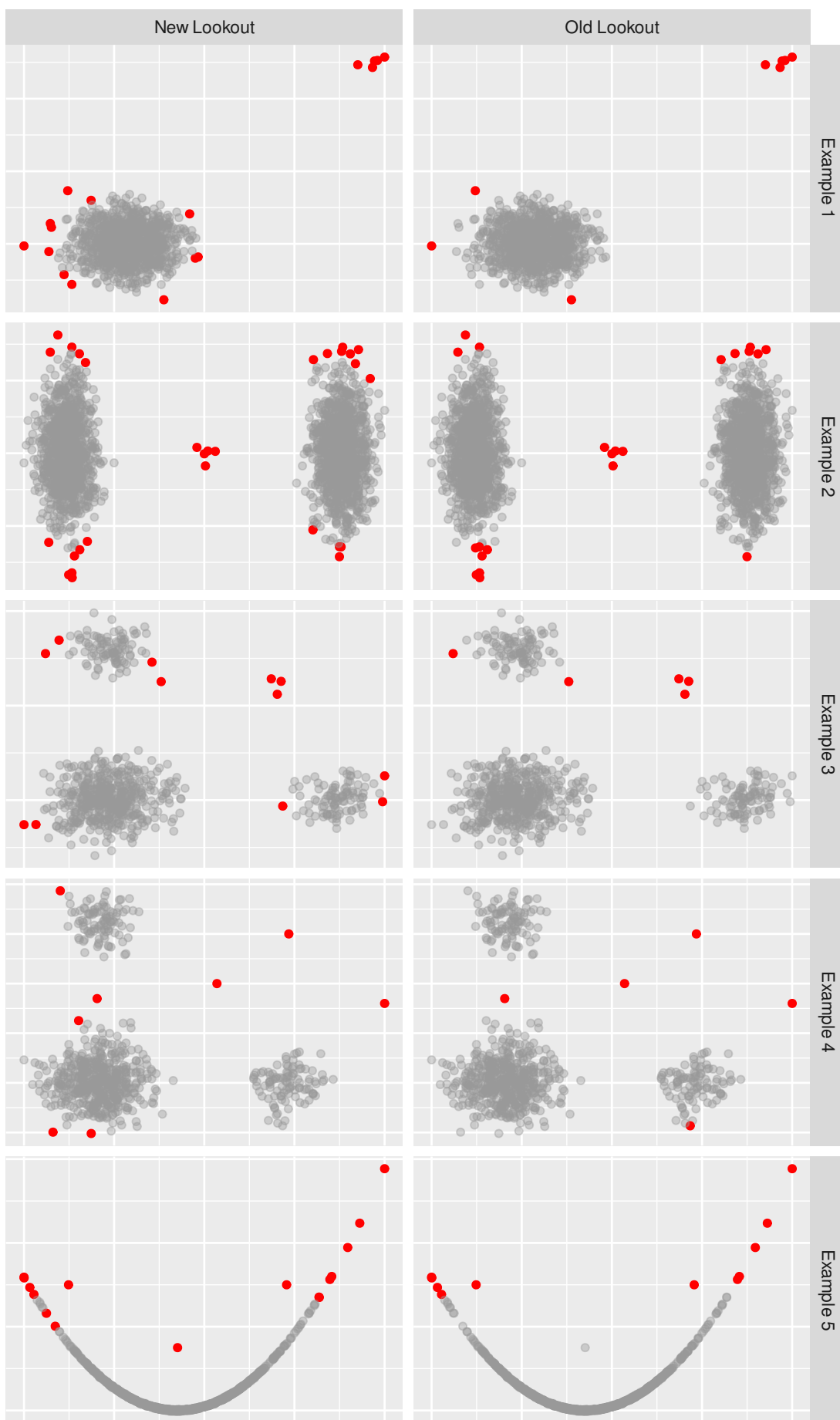


Figure 5: Comparing new lookout with the older version on some showcase examples

5.5 Showcase examples

Fig. 5 shows some examples of new and old versions of lookout with identified anomalies in red. The purpose of this exercise is to check if the new version agrees with the old version on obvious anomalies. The graphs on the first row have anomalies at the top right-hand corner, which are identified by both algorithms; both algorithms find some false anomalies in the lower cloud of points, with the new version finding more. The second row depicts a bi-modal distribution with anomalies placed in the middle. Both versions of lookout identify the central anomalies, along with some false anomalies, with the new version finding more false anomalies. The third row has 3 clusters of normally distributed points with anomalies placed away from the three clusters. Here, the older version of the algorithm finds four false positives compared to ten with the new algorithm. The same setting is repeated in the fourth row with the anomalies spread out, and the two algorithms are similar. The last row has non-anomalous points in a parabola and the anomalous points outside of it. This is a tricky example as the points are on a manifold, and we are not considering their geometry. In this example, only the new algorithm finds all three anomalies lying off the manifold, while both algorithms produce several false positives on the manifold. Overall, on these examples, the new algorithm is more sensitive to identifying anomalies, but also produces a few more false positives. The false anomalies identified by the new algorithm tend to be in regions of the data space with a low density of points, which is consistent with our definition of an anomaly.

5.6 Real-world Examples

We check the new and old versions of lookout on two sets of real data. Neither of these examples have labeled anomalies and so we cannot compute accuracy measures. However, as both datasets have a small number of variables, we can visualize the output. The datasets are

1. Old faithful dataset¹: eruptions of the Old Faithful Geyser in Yellowstone National Park, Wyoming, USA, from 14 January 2017 to 29 December 2023. For each recorded eruption the duration is given in seconds along with the waiting time to the following eruption. The records are incomplete, especially during winter.

¹Data downloaded from <https://geysertimes.org>.

2. Wine reviews dataset²: different attributes of wines from 44 countries, obtained from the *Wine Enthusiast Magazine* during the week of 15 June 2017. We focus on the numerical points score given by the *Wine Enthusiast* reviewer on a scale of 0–100 and the corresponding price of each bottle of wine.



Figure 6: Anomalies in the Old Faithful dataset

Fig. 6 shows the output of new and old versions of lookout when applied to the *Old Faithful* dataset, with the identified anomalies depicted in red. New lookout identifies more “anomalous” points in regions of low density, while the old version tends to identify some points that are close to the main cloud of points as anomalous, and misses some points in the low density regions. In this example, the new lookout algorithm appears to be an improvement over the old.

Fig. 7 shows the output of lookout on the *Wine Reviews* dataset. For this dataset we see that the new version identifies far fewer anomalies compared to the old version, especially at the extremes of the points score, and around the main cloud of points. In this example the new lookout algorithm is a substantial improvement over the old.

²Data downloaded from <https://www.kaggle.com/datasets/zynicide/wine-reviews/data>.

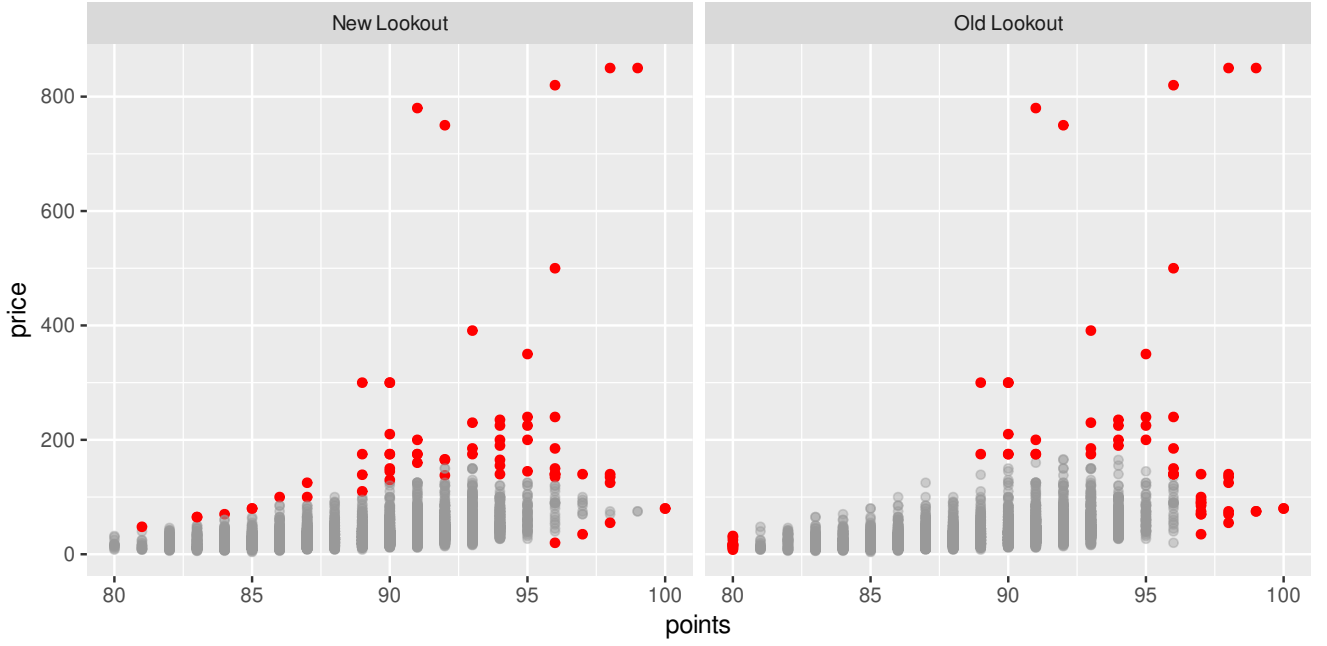


Figure 7: Anomalies in the Wine Reviews dataset. Left with new version of lookout and the right with the old version of lookout.

6 Evaluating lookout with other anomaly detection methods

In this section we compare the new and old versions of lookout with two Extreme Value Theory (EVT) based methods and two KDE-based methods. The EVT-based methods are HDoutliers (Wilkinson 2017) and *stray* (Talagala et al. 2020), both with their α parameter set to zero (which is the default for *stray*). The KDE-based methods are KDEOS (Schubert et al. 2014) and RDOS (Tang & He 2017), both run with their default settings. The same anomaly detection methods were used for comparison in the original version of lookout (Kandanaarachchi & Hyndman 2022).

Both HDoutliers and *stray* identify anomalies, i.e., they give a binary label 1 to anomalies and 0 to non-anomalies. Therefore, to compare lookout with *stray* and HDoutliers we have used the metrics *Fmeasure* and *Gmean*. Suppose TP denotes the true positives (actual = 1, predicted = 1), FP denotes the false positives (actual = 0, predicted = 1), TN denotes the true negatives (actual = 0, predicted = 0) and FN denotes the false negatives (actual = 1, predicted = 0). Then these metrics are defined as follows:

$$Fmeasure = \frac{2(Precision \times Recall)}{Precision + Recall} \quad \text{where} \quad Precision = \frac{TP}{TP + FP} \quad \text{and} \quad Recall = \frac{TP}{TP + FN},$$

$$\text{Gmean} = \sqrt{\text{Sensitivity} \times \text{Specificity}} \quad \text{where} \quad \text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad \text{and} \quad \text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}}.$$

Note that recall and sensitivity define the same quantity. Sensitivity is commonly used in epidemiological settings while recall is used in computer science.

The KDE-based methods KDEOS and RDOS give anomaly scores instead of identifying points as anomalous or not. A standard way to compare anomaly scores is by using the area under the Receiver Operator Curve (ROC). For a given set of anomaly scores the ROC plots $1 - \text{Specificity}$ on the x -axis for different cut-off points and Sensitivity on the y -axis. A method A is better than method B if the area under the ROC is greater for method A compared to B . As lookout gives anomaly scores as well as binary labels, we compare lookout with KDEOS and RDOS using the area under the ROC, which we denote by AUC.

6.1 Experiment 5: Normal distributions moving out in each iteration

In this experiment we consider 405 points in $\mathbb{R}^6$, of which 5 points are anomalies. The experiment is run for 10 iterations with anomalies becoming more obvious in each iteration. The non-anomalies are distributed as $\mathcal{N}(0, 1)$ in all dimensions. The anomalies differ the non-anomalies only in the first dimension. The anomalies are distributed as $\mathcal{N}(2 + (i - 1) \times 0.5, 0.2)$ for iterations $i \in \{1, \dots, 10\}$ in the first dimension. In the other dimensions the anomalies are distributed as $\mathcal{N}(0, 1)$. We run the experiment for 10 iterations and in each iteration the anomalies move further away from the non-anomalous points. Each iteration is repeated 10 times to account for randomness.

Fig. 8 shows the data and the results of this experiment, with anomalies shown in red. The top left plot shows the data on the Y_1Y_2 plane for the 3rd iteration and the middle two plots show the data for the 9th iteration. We see that the anomalies have separated from the other points in the Y_1 direction, but mixed with the non-anomalies on the Y_3Y_4 plane.

The plot on the right gives the AUC when comparing lookout with KDEOS and RDOS and Fmeasure and Gmean for comparison with HDoutliers and stray. As discussed the reason for different metrics is because KDEOS and RDOS give anomaly scores, while stray and HDoutliers identify anomalies. We see that new lookout performs better than the old lookout in terms of Gmean and Fmeasure for higher iterations. In terms of AUC both versions of lookout give the same performance.

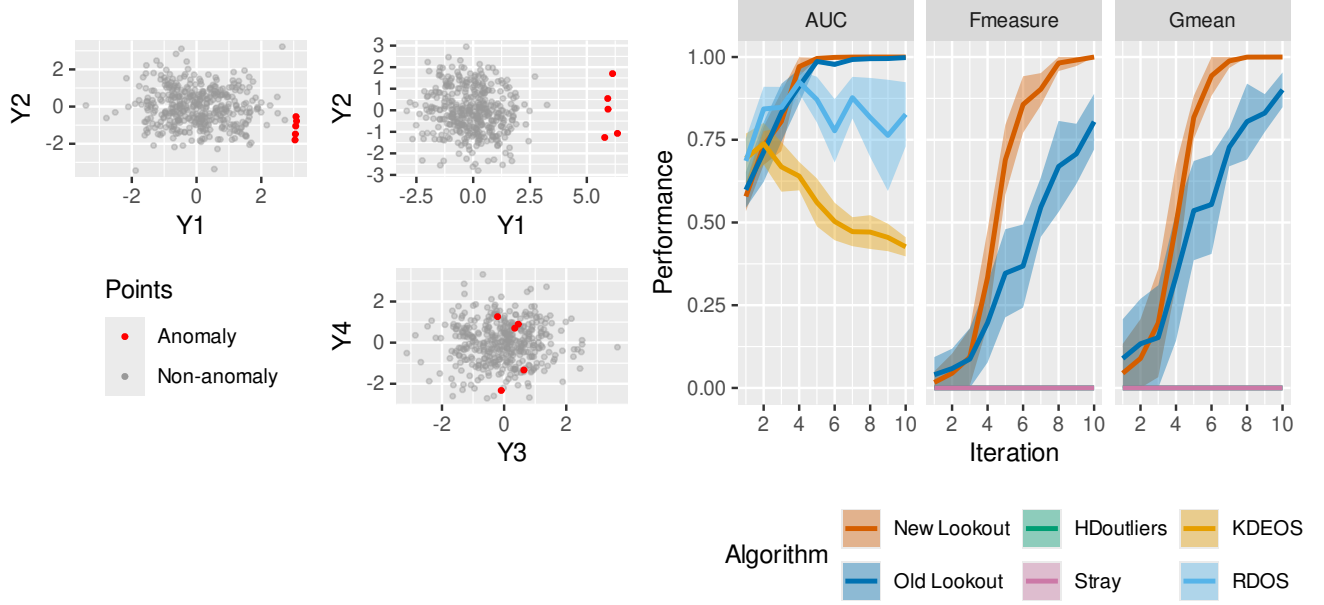


Figure 8: Experiment 5 results. Top left: Data from the 3rd iteration showing the Y_1Y_2 plane with anomalies in red. Middle left: Data from the 9th iteration showing both the Y_1Y_2 plane and Y_3Y_4 plane, with anomalies moving further away in the Y_1 direction, resulting in the anomalies and non-anomalies being mixed on the Y_3Y_4 plane. Right: AUC, Fmeasure and Gmean of old and new versions of lookout, HDoutliers, KDEOS, RDOS and stray.

6.2 Experiment 6: Annulus in $\mathbb{R}^4$

For this experiment we consider 805 points in $\mathbb{R}^3$ of which 5 points are anomalies. Fig. 9 shows the data lying in an annulus on the Y_1Y_2 plane with $Y_3 \sim \mathcal{U}(0, 1)$. As before, the experiment is run for 10 iterations with anomalies moving towards the centre of the hole in each iteration. Plots from iteration 3 and 9 are shown on the left in Fig. 9. The results of the different methods are shown on the right.

6.3 Experiment 7: Unit cube in $\mathbb{R}^{20}$

This experiment considers 500 points in $\mathbb{R}^{20}$ of which one is an anomaly. At the start of the experiment all 500 points are uniformly distributed with $y_i \sim \mathcal{U}(0, 1)$ in each dimension. For this experiment we consider 20 iterations and in each iteration the 500th data point becomes more anomalous. In the first iteration the 500th point is updated with the first coordinate $Y_1 = 0.9$. In the second iteration it is further

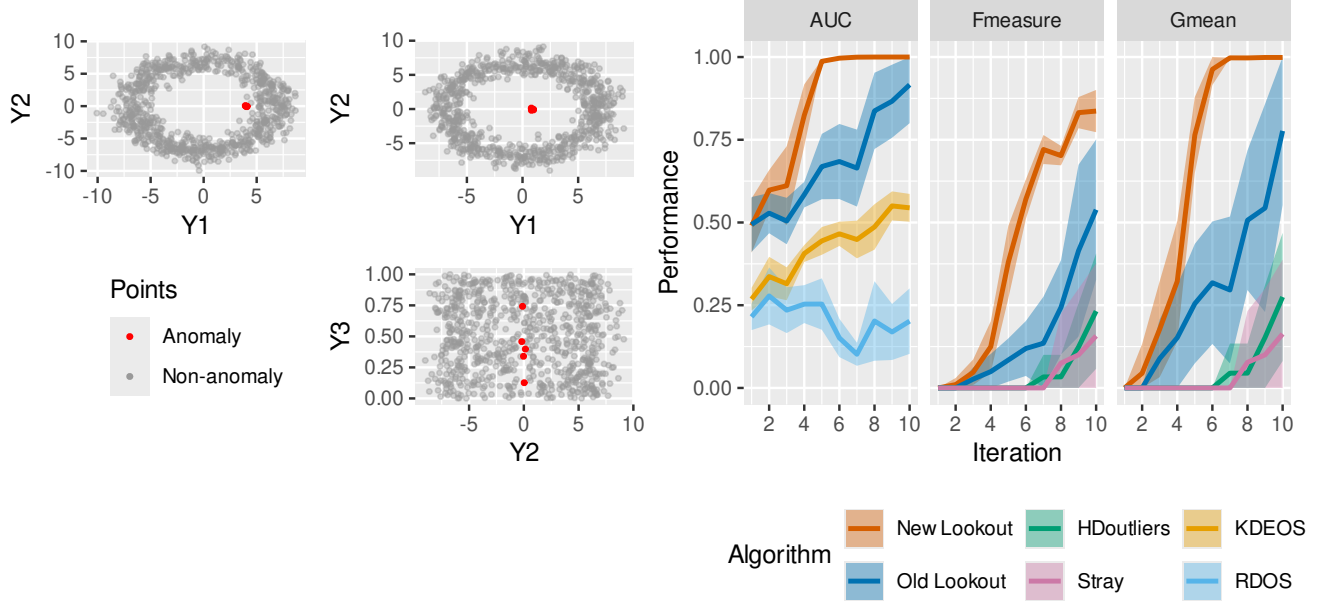


Figure 9: Experiment 6 results. Top left: Data from the 3rd iteration showing the Y_1Y_2 plane with anomalies in red. Middle left: Data from the 9th iteration showing both the Y_1Y_2 plane and Y_2Y_3 plane, with anomalies moving into the annulus. Right: AUC, Fmeasure and Gmean of old and new versions of lookout, HDoutliers, KDEOS, RDOS and stray.

updated with $Y_2 = 0.9$, resulting in $(0.9, 0.9, Y_3, \dots, Y_{20})$. In the i th iteration the 500th point is updated with $Y_i = 0.9$. Each iteration is repeated 10 times to account for randomness.

The anomalous point is not anomalous in any single dimension or on any Y_iY_j plane. In fact even in the space of the first two principal components, the anomaly is not distinguishable. We use *dobin* (Kandanaarachchi & Hyndman 2021), a dimension reduction method especially developed for anomalies to visualize this space. Fig. 10 shows the data and the results of this experiment. The plots on the left show the data on the D_1D_2 plane for iterations 5, 12 and 20, where D_1 and D_2 are the first two axes of *dobin* projection. We see that in the 20th iteration the anomaly can be easily identified on the D_1D_2 plane. The plot on the right shows the results of applying the various algorithms to this problem. The lookout algorithms are applied without scaling in this experiment.

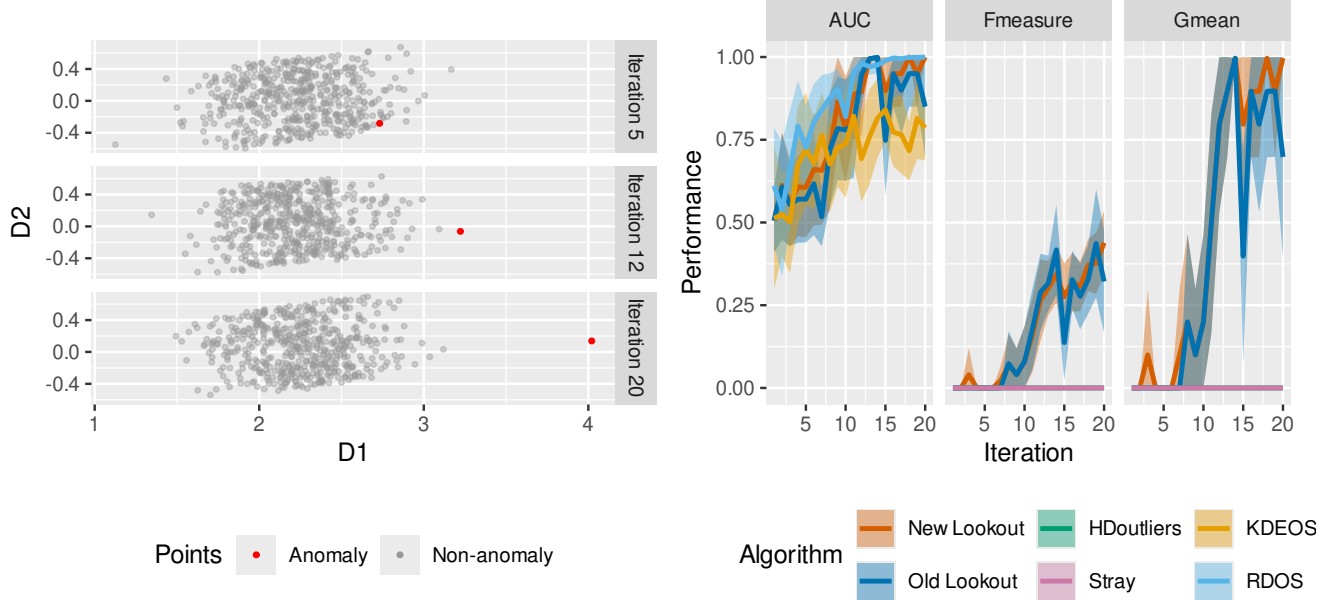


Figure 10: Experiment 7 results. Left: Data from the 5th, 12th and 20th iterations, showing the D_1D_2 plane from the dobin projection. Anomalies are shown in red. Right: AUC, Fmeasure and Gmean of old and new versions of lookout, HDoutliers, KDEOS, RDOS and stray.

7 Discussion and conclusions

We have proposed an improved version of the lookout algorithm based on theoretical underpinnings in persistent homology. The updated algorithm uses an upper quantile of the Rips death diameters rather than the maximum successive difference for bandwidth selection, which we prove yields a consistent kernel density estimator under broad conditions. Additionally, we replace min-max scaling with robust multivariate scaling using the orthogonalized Gnanadesikan-Kettenring estimator, improving robustness to outliers and accounting for correlations between variables. Finally, we constrain the shape parameter of the Generalized Pareto Distribution to be non-positive when fitting to the negative log kernel density estimates, leveraging known bounds on these values to enhance stability.

Our empirical evaluation demonstrates that the updated lookout performs better than the original version across diverse examples, while maintaining competitive performance against other anomaly detection methods. The algorithm provides probabilistic anomaly scores with theoretical grounding in extreme value theory, enabling principled threshold selection.

The key strengths of the updated lookout algorithm include: (1) theoretical consistency guarantees for the kernel density estimator under mild conditions; (2) robustness to outliers in both the scaling and bandwidth selection steps; (3) principled probabilistic interpretation through extreme value theory; and (4) automatic adaptation to the local topological structure of the data through persistent homology.

The algorithm is versatile and suited to any continuous multivariate data, and can handle moderate dimensionality effectively (the experiments show good results up to about 20 dimensions). It can also be applied to any problems that can be transformed into a continuous multivariate setting, such as image anomaly detection via feature extraction, or functional data anomaly detection via functional principal components. We also assume that the data are independent and identically distributed, making it unsuitable for temporal or spatial data without pre-processing (e.g., applying it to the residuals of a fitted model).

However, several computational limitations remain. The computational complexity is dominated by the computation of dimension zero Vietoris-Rips persistent homology, which is equivalent to single link clustering, requiring $O(n \log n)$ time (Gower & Ross 1969). This can be prohibitive for very large datasets, although approximate methods or subsampling could mitigate this. The method may also struggle with very high-dimensional data due to the curse of dimensionality affecting kernel density estimation. Using dimension reduction methods, such as principal component analysis or dobin (Kandanaarachchi 2022), before applying lookout could help in such cases.

A Proofs

Throughout this appendix, b_m denotes the volume of the unit ball in $\mathbb{R}^m$ and $s_{m-1} = mb_m$ the $(m-1)$ -dimensional surface measure of the unit sphere $S^{m-1} \subset \mathbb{R}^m$. For any subset $L \subset \mathbb{R}^m$ and any $\delta > 0$, let L^δ denote the set $\{\mathbf{y} + \mathbf{v} \mid \mathbf{y} \in L, \|\mathbf{v}\| \leq \delta\}$.

A.1 Proof of Theorem 8

The proof rests on a covering estimate for superlevel sets of a Lipschitz density.

Lemma 18. *Let $f : \mathbb{R}^m \rightarrow [0, \infty)$ be a Lipschitz probability density function with Lipschitz constant M ,*

and let $L = f^{-1}[c, \infty)$ be a non-empty superlevel set of f for some $c > 0$. For all $\varepsilon \in (0, c/M]$ there exists $\{\mathbf{x}_1, \dots, \mathbf{x}_l\} \subset L$ such that $L \subseteq \cup_i B(\mathbf{x}_i, \varepsilon)$ with $l \leq 2^m / (cb_m \varepsilon^m)$.

Proof. Fix $\varepsilon \in (0, c/M]$ and take a maximal $(\varepsilon/2)$ -separated subset $\mathcal{S} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_l\}$ within L . This means that $\|\mathbf{x}_i - \mathbf{x}_j\| \geq \varepsilon$ for all $i \neq j$ and that no additional point in L can be added to $\mathcal{S}$ without breaking this condition. By construction this implies that $L \subseteq \cup_i B(\mathbf{x}_i, \varepsilon)$, as otherwise any missed location in L could be added as an additional point and the set remain $(\varepsilon/2)$ -separated.

As each $\mathbf{x}_i \in L$ we also have $B(\mathbf{x}_i, \varepsilon/2) \subset L^{\varepsilon/2}$. Since f is Lipschitz with Lipschitz constant M , we have $L^{\varepsilon/2} \subseteq f^{-1}[c - M\varepsilon/2, \infty)$. For $\varepsilon \leq c/M$ we have $B(\mathbf{x}_i, \varepsilon/2) \subset f^{-1}[c/2, \infty)$, so

$$\mathbb{P}(\mathbf{Y} \in B(\mathbf{x}_i, \varepsilon/2)) = \int_{B(\mathbf{x}_i, \varepsilon/2)} f \, d\mathbf{x} \geq \frac{cb_m \varepsilon^m}{2^m}.$$

Each of the $B(\mathbf{x}_i, \varepsilon/2)$ is disjoint, so $l cb_m \varepsilon^m 2^{-m} \leq \mathbb{P}(\mathbf{Y} \in \cup_i B(\mathbf{x}_i, \varepsilon/2)) \leq 1$. We thus get $l \leq 2^m / (cb_m \varepsilon^m)$. $\square$

Proof of Theorem 8. Choose $c \in (0, 1)$ such that for $L = f^{-1}[c, \infty)$ we have $\mathbb{P}(\mathbf{Y} \notin L) < (1 - \gamma)/3$.

Consider the sequence $r_n := (2^{m+1} / (cb_m n(1 - \gamma)))^{1/m}$. Note that $r_n \in (0, c/M)$ for large enough n ; restrict the sequence to n where this holds. By Lemma 18 there exists a cover of L by l_n balls $\{B(\mathbf{x}_i, r_n) \mid i = 1, 2, \dots, l_n\}$, with

$$l_n \leq \frac{2^m}{cb_m r_n^m} = n(1 - \gamma)/2.$$

Any two points lying in the same ball $B(\mathbf{x}_i, r_n)$ are at distance at most $2r_n$, so they are joined by an edge in the Rips complex at diameter $2r_n$. This implies that the number of connected components at diameter $2r_n$, when restricted to points inside L , is bounded above by $l_n \leq n(1 - \gamma)/2$.

Let $Y(L, n)$ denote the number of the n sample points that are not in L . Then the number of connected components at diameter $2r_n$ is bounded above by $Y(L, n) + n(1 - \gamma)/2$. We have $e_{(\omega_n)} \leq 2r_n$ whenever the complex at diameter $2r_n$ has fewer than $n - \omega_n$ connected components, so it suffices to show that

$$Y(L, n) + n(1 - \gamma)/2 < n - \omega_n$$

for large n . As $n - \omega_n > n(1 - \gamma)$, it is sufficient to show $Y(L, n) < n(1 - \gamma)/2$ for large n .

From the construction of L , $Y(L, n)$ is binomial with mean np and variance $np(1-p)$, with $p < (1-\gamma)/3$. Chebyshev's inequality gives

$$\begin{aligned}
\mathbb{P}(Y(L, n) > n(1-\gamma)/2) &\leq \mathbb{P}(|Y(L, n) - np| > n(1-\gamma)/2 - n(1-\gamma)/3) \\
&= \mathbb{P}(|Y(L, n) - np| > n(1-\gamma)/6) \\
&\leq \frac{\text{Var}(Y(L, n))}{(n(1-\gamma)/6)^2} \\
&\leq \frac{36np(1-p)}{n^2(1-\gamma)^2} \\
&\leq \frac{12}{n(1-\gamma)}.
\end{aligned}$$

If there are fewer than $n(1-\gamma)$ connected components at diameter $2r_n$ then $e_{(\omega_n)} \leq 2r_n =: C(f, \gamma)n^{-1/m}$, where $C(f, \gamma) = 2(2^{m+1}/(cb_m(1-\gamma)))^{1/m}$. Thus $\mathbb{P}(e_{(\omega_n)} > C(f, \gamma)n^{-1/m}) \leq 12/(n(1-\gamma))$, with $C(f, \gamma)$ dependent on f and γ but independent of n . $\square$

A.2 Proof of Counterexample 14

For the remainder of this appendix we consider the distribution on $\mathbb{R}^m$, $m \geq 3$, with density $f(\mathbf{x}) = c_m/(1 + \|\mathbf{x}\|^{m+1})$, where c_m is the appropriate normalising constant. A useful quantity is the number of samples with radius at least R : let $A_n^{[R, \infty)}$ be the number of samples lying in $\{\mathbf{x} \in \mathbb{R}^m \mid \|\mathbf{x}\| \geq R\}$.

We want to show that outside a certain radius R_n all the points are isolated at unit length scale and thus each contribute their own connected component. A sufficient condition is that there be no pair of points within distance 1 of each other among those of radius at least $R_n - 1$. We claim nothing about points in the annulus from radius $R_n - 1$ to R_n , which could be joined to points of smaller radius.

Lemma 19. *Fix a sequence of positive numbers R_n . Let $\mathcal{S}_n$ be a set of n points sampled i.i.d. according to density f , and let $g(n)$ be any function bounded above by $\sqrt{R_n^{m-1}/\log n}$. Let Y_n be the number of pairs within distance 1 of each other among the points in $\{\mathbf{x} \in \mathbb{R}^m \mid \|\mathbf{x}\| > R_n - 1\}$. Then*

$$\mathbb{P}\left(Y_n = 0 \mid A_n^{[R_n-1, \infty)} < \sqrt{\frac{R_n^{m-1}}{\log n}} \text{ and } A_n^{[R_n, \infty)} > g(n)\right) \rightarrow 1$$

as $n \rightarrow \infty$.

Proof. We first bound the probability that two points of large radius are within distance 1:

$$\mathbb{P}(\|\mathbf{x} - \mathbf{y}\| \leq 1 \mid \|\mathbf{x}\|, \|\mathbf{y}\| \geq R - 1) = \frac{1}{\mathbb{P}(\|\mathbf{x}\|, \|\mathbf{y}\| \geq R - 1)} \int_{R-1}^{\infty} \frac{c_m s_{m-1} r^{m-1}}{1 + r^{m+1}} \mathbb{P}(\mathbf{y} \in B(\mathbf{x}, 1) \mid \|\mathbf{x}\| = r) dr.$$

If $\|\mathbf{x}\| = r$ with r sufficiently large, then the probability that another sample $\mathbf{y}$ lies in $B(\mathbf{x}, 1)$ is bounded above by $\int_{B(\mathbf{x}, 1)} c_m / (1 + \|\mathbf{z}\|^{m+1}) d\mathbf{z} \leq C/r^m$ for some constant C independent of r . This implies that there exists a constant C_1 independent of R such that

$$\mathbb{P}(\|\mathbf{x} - \mathbf{y}\| \leq 1 \mid \|\mathbf{x}\|, \|\mathbf{y}\| > R - 1) < (R - 1)^2 C_1 \int_{R-1}^{\infty} \frac{1}{r^{m+2}} dr,$$

and hence a constant C_2 independent of R , for sufficiently large R , such that

$$\mathbb{P}(\|\mathbf{x} - \mathbf{y}\| \leq 1 \mid \|\mathbf{x}\|, \|\mathbf{y}\| > R - 1) \leq \frac{C_2}{R^{m-1}}.$$

For each pair of points the probability of being within distance 1 is thus bounded above by C_2/R_n^{m-1} .

Changing the function $g(n)$ does not invalidate this upper bound. Together we get

$$\mathbb{E} \left[Y_n \mid A_n^{[R_n-1, \infty)} \leq \sqrt{\frac{R_n^{m-1}}{\log n}} \text{ and } A_n^{[R_n, \infty)} > g(n) \right] \leq \left(\frac{R_n^{m-1}}{\log n} \right) \frac{C_2}{2R_n^{m-1}} = \frac{C_2}{2 \log n}$$

for some C_2 independent of sufficiently large n . Since Y_n is a non-negative integer, $\mathbb{P}(Y_n \neq 0 \mid \cdot) \leq \mathbb{E}[Y_n \mid \cdot] \rightarrow 0$, and the result follows. $\square$

Given the above lemma, our next step is to show that with positive probability there are enough points of radius at least R_n , but not too many of radius more than $R_n - 1$.

Lemma 20. *Let $\mathcal{S}_n$ be a sample of points i.i.d. according to f , and set $R_n = n^{2/m}/(c_m s_{m-1})$. Then*

$$\mathbb{P} \left(A_n^{[R_n-1, \infty)} < 3n^{1-2/m} \text{ and } A_n^{[R_n, \infty)} > \frac{1}{4}n^{1-2/m} \right) > 1/3$$

for n sufficiently large.

Proof. For each radius R , the probability that a sample point has modulus at least R is

$$p_{[R, \infty)} = \int_R^{\infty} \frac{c_m s_{m-1} r^{m-1}}{1 + r^{m+1}} dr.$$

We have $p_{[R, \infty)} \in (c_m s_{m-1}/(2R), c_m s_{m-1}/R)$ for $R > 1$. Thus $A_n^{[R_n, \infty)} \sim \mathcal{B}(n, p)$, the binomial distribution, for some $p \in (n^{-2/m}/2, n^{-2/m})$. Using Chebyshev's inequality,

$$\mathbb{P} \left(A_n^{[R_n, \infty)} \in \left(\frac{1}{4}n^{1-2/m}, 2n^{1-2/m} \right) \right) > \frac{1}{2}.$$

Given that a sample point $\mathbf{x}$ lies in $B(\mathbf{0}, R)$, the probability that $\mathbf{x}$ has modulus in $[R - 1, R)$ is

$$\hat{p} = \frac{\int_{R-1}^R \frac{c_m s_{m-1} r^{m-1}}{1 + r^{m+1}} dr}{1 - p_{[R, \infty)}} < \frac{-\frac{c_m s_{m-1}}{R} + \frac{c_m s_{m-1}}{R-1}}{1 - \frac{c_m s_{m-1}}{R}} < \frac{2c_m s_{m-1}}{R^2}$$

for sufficiently large R . For $R_n = 2n^{2/m}/(c_m s_{m-1})$ we have $\hat{p}_n \leq C_1 n^{-4/m}$ for large n , with C_1 a constant independent of n .

Let $\hat{A}_n^{[R_n-1, R_n]}$ be the number of samples with $\|\mathbf{x}\| \in [R_n - 1, R_n]$, conditioned on there being $A_n^{[R_n, \infty)}$ samples outside $B(\mathbf{0}, R_n)$. Then $\hat{A}_n^{[R_n-1, R_n]} \sim \mathcal{B}(n - A_n^{[R_n, \infty)}, \hat{p}_n)$. Coupling $\mathcal{B}(n - A_n^{[R_n, \infty)}, \hat{p}_n)$ to $\mathcal{B}(n, \hat{p}_n)$ implies that $\mathbb{P}(\hat{A}_n^{[R_n-1, R_n]} \leq n^{1-2/m}) > 2/3$ for large enough n . Combining,

$$\begin{aligned} & \mathbb{P}\left(A_n^{[R_n-1, \infty)} < 3n^{1-2/m} \text{ and } A_n^{[R_n, \infty)} > \frac{1}{4}n^{1-2/m}\right) \\ & \geq \mathbb{P}\left(A_n^{[R_n-1, \infty)} < 3n^{1-2/m} \text{ and } A_n^{[R_n, \infty)} \in \left(\frac{1}{4}n^{1-2/m}, 2n^{1-2/m}\right)\right) \\ & \geq \mathbb{P}\left(\hat{A}_n^{[R_n-1, R_n]} < n^{1-2/m} \mid A_n^{[R_n, \infty)} \in \left(\frac{1}{4}n^{1-2/m}, 2n^{1-2/m}\right)\right) \mathbb{P}\left(A_n^{[R_n, \infty)} \in \left(\frac{1}{4}n^{1-2/m}, 2n^{1-2/m}\right)\right) \\ & \geq \frac{2}{3} \cdot \frac{1}{2} = \frac{1}{3}. \end{aligned} \quad \square$$

We can finally combine these to complete the counterexample. The probabilities are far from tight, as we need only show that $e_{(\omega_n)}$ does not converge to 0.

Proof of [Counterexample 14](#). A sufficient condition for $e_{(\omega_n)} \geq 1$ is that the graph joining points within distance 1 of each other has $\frac{1}{4}n^{1-2/m}$ isolated points. One way to obtain this many isolated points is to have $A_n^{[R_n, \infty)} > \frac{1}{4}n^{1-2/m}$ and $Y_n = 0$. Consider $R_n = n^{2/m}/(c_m s_{m-1})$.

From [Lemma 20](#) we have $\mathbb{P}(A_n^{[R_n-1, \infty)} < 3n^{1-2/m} \text{ and } A_n^{[R_n, \infty)} > \frac{1}{4}n^{1-2/m}) > 1/3$ for n sufficiently large. Substituting in R_n shows that $3n^{1-2/m} < \sqrt{R_n^{m-1}/\log n}$, so [Lemma 19](#) applies and gives $\mathbb{P}(Y_n = 0 \mid A_n^{[R_n-1, \infty)} < 3n^{1-2/m} \text{ and } A_n^{[R_n, \infty)} > \frac{1}{4}n^{1-2/m}) \rightarrow 1$ as $n \rightarrow \infty$, hence at least $\frac{3}{4}$ for sufficiently large n . Combining,

$$\begin{aligned} \mathbb{P}(e_{(\omega_n)} > 1) & \geq \mathbb{P}\left(Y_n = 0 \text{ and } A_n^{[R_n-1, \infty)} < 3n^{1-2/m} \text{ and } A_n^{[R_n, \infty)} > \frac{1}{4}n^{1-2/m}\right) \\ & \geq \frac{1}{3} \cdot \frac{3}{4} = \frac{1}{4}. \end{aligned} \quad \square$$

Disclosure statement

There are no conflicts of interest to declare.

Supplementary material

The updated algorithm is implemented in the `lookout` R package (Kandanaarachchi et al. 2025), available on CRAN at <https://CRAN.R-project.org/package=lookout>.

Files to reproduce all numerical examples are available at <https://github.com/robjhyndman/look-out-2>

References

- Adler, R. J., Bobrowski, O. & Weinberger, S. (2014), ‘Crackle: The homology of noise’, *Discrete & computational geometry* **52**, 680–704.
URL: <https://doi.org/10.1007/s00454-014-9621-6>
- Barnett, V. & Lewis, T. (1978), *Outliers in Statistical data*, John Wiley & Sons, Chichester, UK.
- Breunig, M. M., Kriegel, H.-P., Ng, R. T. & Sander, J. (2000), ‘LOF: identifying density-based local outliers’, *ACM Sigmod Record* **29**(2), 93–104.
- Chacón, J. E. & Duong, T. (2018), *Multivariate kernel smoothing and its applications*, CRC Press, Boca Raton, Florida.
- Coles, S. (2001), *An introduction to statistical modeling of extreme values*, Vol. 208 of *Springer Series in Statistics*, Springer, London UK.
- Fraley, C. (2018), *HDoutliers: Leland Wilkinson’s Algorithm for Detecting Multidimensional Outliers*. R package v1.0.
URL: <https://CRAN.R-project.org/package=HDoutliers>
- Friedman, J. H. & Rafsky, L. C. (1979), ‘Multivariate generalizations of the Wald-Wolfowitz and Smirnov two-sample tests’, *The Annals of Statistics* **7**(4), 697–717.
- Ghrist, R. (2008), ‘Barcodes: the persistent topology of data’, *Bulletin of the American Mathematical Society* **45**(1), 61–75.

- Good, I. J. (1952), ‘Rational decisions’, *Journal of the Royal Statistical Society. Series B, Statistical methodology* **14**, 107–114.
- URL:** <http://dx.doi.org/10.1111/j.2517-6161.1952.tb00104.x>
- Gower, J. C. & Ross, G. J. S. (1969), ‘Minimum spanning trees and single linkage cluster analysis’, *Journal of the Royal Statistical Society. Series C, Applied Statistics* **18**, 54–64.
- Hawkins, D. M. (1980), *Identification of outliers*, Chapman & Hall, London, UK.
- Hyndman, R. J. & Fan, Y. (1996), ‘Sample quantiles in statistical packages’, *The American Statistician* **50**, 361–365.
- Hyndman, R. J. & Frazier, D. T. (2026), Anomaly detection using surprisals, Working paper, Department of Econometrics & Business Statistics, Monash University.
- URL:** <http://robjhyndman.com/publications/surprisals.html>
- Kandanaarachchi, S. (2022), *dobin: Dimension Reduction for Outlier Detection*. R package v1.0.4.
- URL:** <https://CRAN.R-project.org/package=dobin>
- Kandanaarachchi, S. & Hyndman, R. J. (2021), ‘Dimension reduction for outlier detection using DOBIN’, *Journal of Computational and Graphical Statistics* **30**(1), 204–219.
- URL:** <https://doi.org/10.1080/10618600.2020.1807353>
- Kandanaarachchi, S. & Hyndman, R. J. (2022), ‘Leave-one-out kernel density estimates for outlier detection’, *Journal of Computational and Graphical Statistics* **31**(2), 586–599.
- Kandanaarachchi, S., Hyndman, R. J. & Fraley, C. (2025), *lookout: Leave One Out Kernel Density Estimates for Outlier Detection*. R package v2.0.1.
- URL:** <https://sevvandi.github.io/lookout/>
- Kolupuri, S. V. J., Paul, A., Bhowmick, R. S. & Ganguli, I. (2025), Scams and frauds in the digital age: ML-based detection and prevention strategies, in ‘Proceedings of the 26th International Conference on Distributed Computing and Networking’, pp. 340–345.

- March, W. B., Ram, P. & Gray, A. G. (2010), Fast Euclidean minimum spanning tree: algorithm, analysis, and applications, *in* ‘Proceedings of the 16th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining’, ACM, New York, pp. 603–612.
- Penrose, M. D. (1997), ‘The longest edge of the random minimal spanning tree’, *The Annals of Applied Probability* **7**(2), 340–361.
- Penrose, M. D. (1998), ‘Extremes for the minimal spanning tree on normally distributed points’, *Advances in Applied Probability* **30**(3), 628–639.
- Penrose, M. D. & Yukich, J. E. (2003), ‘Weak laws of large numbers in geometric probability’, *The Annals of Applied Probability* **13**(1), 277–303.
- Pickands, J. (1975), ‘Statistical inference using extreme order statistics’, *The Annals of Statistics* **3**(1), 119–131.
- Rohlf, F. J. (1975), ‘Generalization of the gap test for the detection of multivariate outliers’, *Biometrics* **31**(1), 93–101.
- Rousseeuw, P. J. & Driessen, K. V. (1999), ‘A fast algorithm for the minimum covariance determinant estimator’, *Technometrics* **41**(3), 212–223.
- Schubert, E., Zimek, A. & Kriegel, H. P. (2014), ‘Generalized outlier detection with flexible kernel density estimates’, *SIAM International Conference on Data Mining 2014* pp. 542–550.
- Singh Parihar, Y., Curtin, R., Eddelbuettel, D. & Balamuta, J. (2026), *mlpack: Rcpp Integration for the mpack Library*. R package version 4.8.0-1.
URL: <https://www.mpack.org/doc/user/bindings/r.html>
- Steele, J. M. (1988), ‘Growth rates of Euclidean minimal spanning trees with power weighted edges’, *The Annals of Probability* **16**(4), 1767–1787.
- Stone, J. V. (2022), *Information theory: a tutorial introduction*, Sebtel Press, Sheffield, England.
- Talagala, P. D., Hyndman, R. J. & Smith-Miles, K. (2021), ‘Anomaly detection in high-dimensional data’, *Journal of Computational and Graphical Statistics* **30**(2), 360–374.

- Talagala, P. D., Hyndman, R. J., Smith-Miles, K., Kandanaarachchi, S. & Muñoz, M. A. (2020), ‘Anomaly detection in streaming nonstationary temporal data’, *Journal of Computational and Graphical Statistics* **29**(1), 13–27.
- Tang, B. & He, H. (2017), ‘A local density-based approach for outlier detection’, *Neurocomputing* **241**, 171–180.
- Wilkinson, L. (2017), ‘Visualizing big data outliers through distributed aggregation’, *IEEE Transactions on Visualization and Computer Graphics* **24**(1), 256–266.
- Yukich, J. E. (1998), *Probability Theory of Classical Euclidean Optimization Problems*, Vol. 1675 of *Lecture Notes in Mathematics*, Springer, Berlin.